7

The Regulation of Calcium, Magnesium, and Phosphate Excretion by the Kidney

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CHAPTER OUTLINE

KEY POINTS

- $1\alpha,25(OH)_2D_3$, parathyroid hormone (PTH), and the phosphatonin, FGF-23, regulate renal tubule function to maintain homeostasis of serum calcium and phosphate concentrations, whereas magnesium handling is regulated largely by dietary magnesium load.
- The hypocalciuric effect of PTH is mediated by increased paracellular Ca_2^+ reabsorption in the thick ascending limb of Henle, via inhibition of claudin-14 and by changes in the expression of apical channels in the distal tubule.
- Sclerostin is an osteocyte-derived glycoprotein that has calciuric effects and appears to serve as a counterbalance to PTH and $1\alpha,25(OH)_2$ in calcium homeostasis.
- Claudin-10b is a tight junction protein that mediates paracellular Na^+ reabsorption in the medullary thick ascending limb, thereby limiting Ca_2^+ reabsorption further downstream in the cortical thick ascending limb.
- Mutations in claudin-10b cause hypermagnesemia and hypokalemic alkalosis, together with various extrarenal manifestations.
- PTH and FGF-23 converge on a common signaling pathway in the proximal tubule, involving phosphorylation of NHERF-1 and dissociation and internalization of the NaPi-IIa cotransporter, to cause phosphaturia.
- $1\alpha,25(OH)_2D$ stimulates the synthesis of FGF-23, which in turn inhibits 25-hydroxyvitamin D 1α-hydroxylase, serving as a negative feedback loop to limit vitamin D effects on phosphate turnover.

CALCIUM TRANSPORT IN THE KIDNEY

THE ROLE OF CALCIUM IN CELLULAR PROCESSES

Calcium is an abundant cation in the body (Table 7.1). Several biochemical and physiologic processes, including nerve conduction and function, coagulation, enzyme activity, exocytosis, and bone mineralization, are critically dependent on normal calcium concentrations in extracellular fluid.[1–3] Not unexpectedly, significant decreases or increases in serum calcium concentrations are associated with marked symptoms and signs. Intricate mechanisms exist to maintain extracellular fluid calcium concentrations within a narrow range and to maintain calcium balance. Decreases in serum calcium concentrations are associated with numbness and tingling of the extremities and peri-oral region, cramping, Chvostek and Trousseau signs, tetany, and when profound, generalized seizures.[4–6] A negative calcium balance that is present when calcium absorption in the intestine is reduced is associated with secondary hyperparathyroidism, hypophosphatemia, and rickets or osteomalacia.[5,6] Hypercalcemia, especially when severe, is associated with lethargy, confusion, irritability, depression, hallucinations, and in extreme cases, stupor and coma, anorexia, nausea, vomiting and constipation, cardiac ectopy,

Table 7.1 Composition of the Whole Body as Determined by Chemical Analysis (Values per Kilogram Fat-Free Tissue Unless Otherwise Indicated)

Body Weight (kg)	Water[a] (g)	Fat[a] (g)	Water (g)	N (g)	Na (mEq)	K (mEq)	Cl (mEq)	Mg (g)	Ca (g)	P (g)	Fe (mg)	Cu (mg)	Zn (mg)	B (mg)	Co (mg)
70	605	160	720	34	80	69	50	0.47	22.4	12.0	74	1.7	28	0.37	0.02

[a]Per kilogram whole body weight.

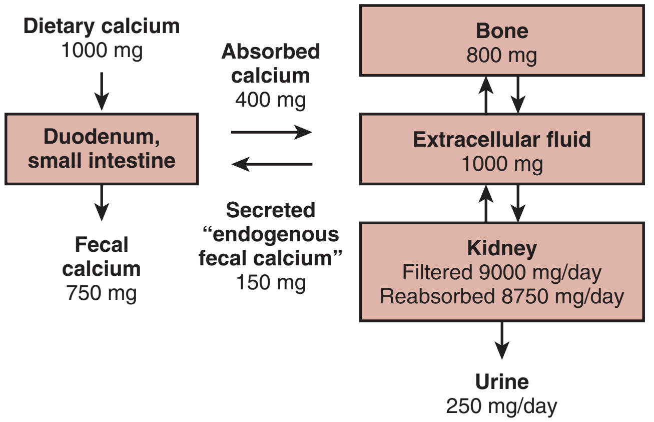


Fig. 7.1 Calcium homeostasis in normal humans showing the amounts of calcium absorbed in the intestine and reabsorbed by the kidney.

and polyuria and renal colic from the passage of renal stones.[7] The attendant hypercalciuria is associated with a reduced capacity to concentrate urine,[8–11] volume depletion, and nephrocalcinosis and renal stones.[12,13] Hypercalcemia occurs as a result of parathyroid hormone (PTH)-dependent or PTH-independent processes, and changes in laboratory values depend on the etiology of hypercalcemia.[7] Thus, in PTH-dependent hypercalcemia, elevated serum PTH concentrations are present and are the cause of hypercalcemia, whereas in PTH-independent hypercalcemia, PTH concentrations are suppressed, sometimes in association with increases in various vitamin D metabolites.[7] As shown in Fig. 7.1, the intestine and kidney are important in the absorption and the reabsorption and excretion of calcium. Following the absorption in the intestine, calcium in the extracellular fluid space is deposited in bone (the major repository of calcium in the body) and is filtered in the kidney. The concentration of calcium in serum varies with age and gender, with higher values present in children and adolescent subjects than in adults.

CALCIUM IS PRESENT IN SERUM IN BOUND AND FREE FORMS

Calcium is present in plasma in filterable (60% of total calcium) and bound (40% of total calcium) forms. Filterable calcium is composed of calcium complexed to anions, such as citrate, sulfate, and phosphate (~10% of total calcium) and ionized calcium (~50% of total calcium) (Fig. 7.2).[14] The percentage of calcium bound to proteins (predominantly albumin, and to a lesser extent, globulins), and the amount of filterable calcium, is dependent on plasma pH.[14] Alkalemia is associated with a reduction in free calcium, whereas acidemia is associated with

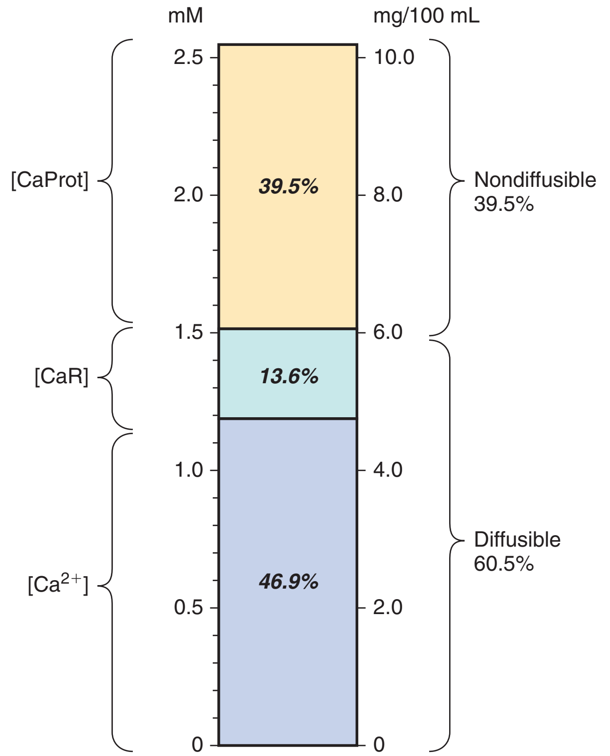


Fig. 7.2 Components of serum total calcium assessed by ultrafiltration data in normal human patients. *CaR,* Diffuse of both calcium complexes; *Ca^{2+},* ionized calcium, *CaProt,* protein-bound calcium. Redrawn from Moore, EW. Ionized calcium in normal serum, ultrafiltrates, and whole blood determined by ion-exchange electrodes. *J Clin Invest*. 1970;49:318–334, with permission of the publisher.

an increase in free calcium. A 1-g/dL change in serum albumin is associated with a 0.8-mg/dL change in total serum calcium, and a 1-g/dL change in globulins is associated with a 0.16-mg/dL change in total serum calcium. An equation defining the amount of calcium (mmol/L) bound to albumin and globulins (g/L) as a function of pH is as follows[14]:

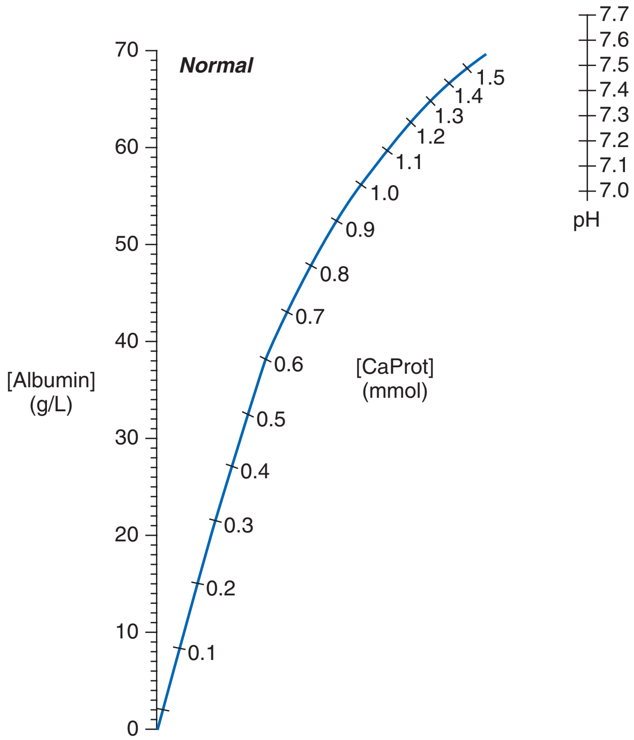
$$[CaProt] = 0.019[Alb] - [(0.42)([Alb]/47.3)(7.42 - pH)] + 0.004[Glob] - [(0.42)([Glob]/25.0)(7.42 - pH)]$$

If one assumes that all calcium is bound to albumin, the following equation applies:

$$[CaProt] = 0.0211[Alb] - [(0.42)([Alb]/47.3)(7.42 - pH)],$$

where Alb is albumin, CaProt is protein-bound calcium, and Glob is globulins.

A nomogram describing this relationship is shown in Fig. 7.3.



$$[\text{CaProt}] = [0.0211[\text{Alb}]] - [(0.42)\frac{[\text{Alb}]}{47.3}(7.42 - \text{pH})]$$

Fig. 7.3 A nomogram for estimating protein-bound calcium levels [CaProt] in normal humans. [CaProt] is obtained by connecting observed albumin and pH values with a straight line and reading the point at which it intersects the curve. The equation describing the relationship between [CaProt] (protein-bound calcium), serum albumin concentrations, and pH is shown at the bottom of the graph. Redrawn from Moore EW. Ionized calcium in normal serum, ultrafiltrates, and whole blood determined by ion-exchange electrodes. *J Clin Invest*. 1970;49:318–334, with permission of the publisher.

REGULATION OF CALCIUM HOMEOSTASIS BY THE PARATHYROID HORMONE–VITAMIN D ENDOCRINE SYSTEM

In states of neutral calcium balance, the amount of calcium absorbed by the intestine is equivalent to the amount excreted by the kidney. The central role of the vitamin D–PTH endocrine system in the regulation of calcium homeostasis is well recognized.[15–17] The major physiologic role of vitamin D through the activity of its active metabolite 1α,25-dihydroxyvitamin D_3 (1α,25$(OH)_2D_3$) is the maintenance of normal calcium and phosphorus balance.[18–21] In Fig. 7.4 the metabolism of vitamin D and the adaptations that occur in response to changes in serum calcium are illustrated. Fig. 7.4A summarizes the salient biochemical transformations that occur endogenously during the formation and metabolism of vitamin D_3. Vitamin D_3 (cholecalciferol) is formed in the skin by the ultraviolet light–mediated photolysis of the B-ring of the sterol precursor, 7-dehydrocholesterol, which gives rise to pre-vitamin D_3 that rapidly undergoes thermal equilibration to vitamin D_3.[22–30] Vitamin D_2, or ergocalciferol, derived by photolysis of the plant sterol, ergosterol, is ingested orally, after which it is metabolized in a similar manner to vitamin D_3.[31] Although vitamin D_2 is less active in birds than mammals, when compared with vitamin D_3, the metabolic transformations of vitamin D_2 and vitamin D_3 are similar.[32] Vitamin D_3, bound to vitamin D–binding protein, to which it preferentially binds relative to its precursor, pre-vitamin D_3,[29] exits the skin, enters the circulation, and is metabolized in the liver microsomes and mitochondria to 25-hydroxyvitamin D_3 (25(OH)D_3) by the vitamin D_3-25-hydroxylase.[16,33–43] The CYP2R1 is the cytochrome P450 of the microsomal vitamin D_3-25 hydroxylase.[41] Other vitamin D_3-25-hydroxylases play a role in the transformation of vitamin D_3 to 25(OH)D_3 because deletion of the *Cyp2r1* gene in mice results in reduced (>50% reduction) but detectable serum 25(OH)D_3 concentrations.[43]

The subsequent metabolism of 25(OH)D_3 is dependent on the calcium and phosphorus requirements of the individual. In states of calcium demand, 25(OH)D_3 is metabolized by the 25-hydroxyvitamin D_3-1α-hydroxylase to the biologically active vitamin D metabolite, 1α,25-dihydroxyvitamin D_3 (1α,25$(OH)_2D_3$), in mitochondria of kidney proximal and distal tubule cells by PTH-dependent processes (see Fig. 7.4B).[18,26,44–56] In response to reductions in calcium intake and subsequent decreases in serum calcium, PTH release from the parathyroid glands is increased. The change in serum calcium concentrations is detected by the parathyroid gland calcium-sensing receptor, a G-protein–coupled receptor, which alters PTH release from the parathyroid cell.[57–60] PTH enhances calcium transport in the distal tubule of the kidney directly,[61–63] and indirectly through changes in sclerostin expression,[55,56,64,65] and increases the activity of the renal 25-hydroxyvitamin D 1α-hydroxylase, and attendant increases in the synthesis of 1α,25$(OH)_2D_3$.[52] 1α,25$(OH)_2D_3$ increases calcium transport in the intestine[49,50,66] and kidney.[67–71] At the same time, both PTH[72–74] and 1α,25$(OH)_2$D[75,76] increase bone calcium mobilization and help to maintain serum calcium concentrations. The converse series of events occurs in hypercalcemic circumstances. In states of calcium sufficiency, the synthesis of 1α,25$(OH)_2D_3$ is reduced, and the synthesis of 24*R*,25-dihydroxyvitamin D_3 (24*R*,25$(OH)_2D_3$), a vitamin D metabolite with reduced bioactivity, is increased.[77–79] The synthesis of 24*R*,25$(OH)_2D_3$ is mediated by a 1α,25$(OH)_2D_3$-inducible enzyme, the 25(OH) D_3-24-hydroxylase, that is present in several target tissues of 1α,25$(OH)_2D_3$ including the intestine and the kidney.[77,80–84]

1α,25$(OH)_2D_3$, PTH, and the phosphatonin, fibroblast growth factor-23 (FGF-23), regulate and maintain normal phosphorus concentrations.[85–87] Serum phosphate concentrations also regulate the synthesis of 1α,25$(OH)_2D_3$ by PTH independent mechanisms.[88] In states of phosphorus demand, 25(OH)D_3 is metabolized to 1α,25$(OH)_2D_3$ and the synthesis of 24*R*,25$(OH)_2D_3$ is reduced.[16,46,89–92] The enhanced synthesis of 1α,25$(OH)_2D_3$ in hypophosphatemic states is induced directly by reductions in serum phosphate,[88,93–95] an increase in the expression of IGF-1,[96,97] and by inhibition of FGF-23.[98] The converse occurs in hyperphosphatemic states. A decrease in serum phosphate concentrations is associated with an increase in ionized calcium, a decrease in PTH secretion, and a subsequent decrease in renal phosphate excretion. An increase in renal 25-hydroxyvitamin D-1α-hydroxylase activity, increased 1α,25$(OH)_2D_3$ synthesis, and increased phosphorus absorption in the intestine and reabsorption in the kidney occur.[88,92,94,99–105] In the intestine and kidney, 1α,25$(OH)_2D_3$ increases the expression of the sodium–phosphate cotransporters IIb, and IIa and IIc, respectively, thereby regulating

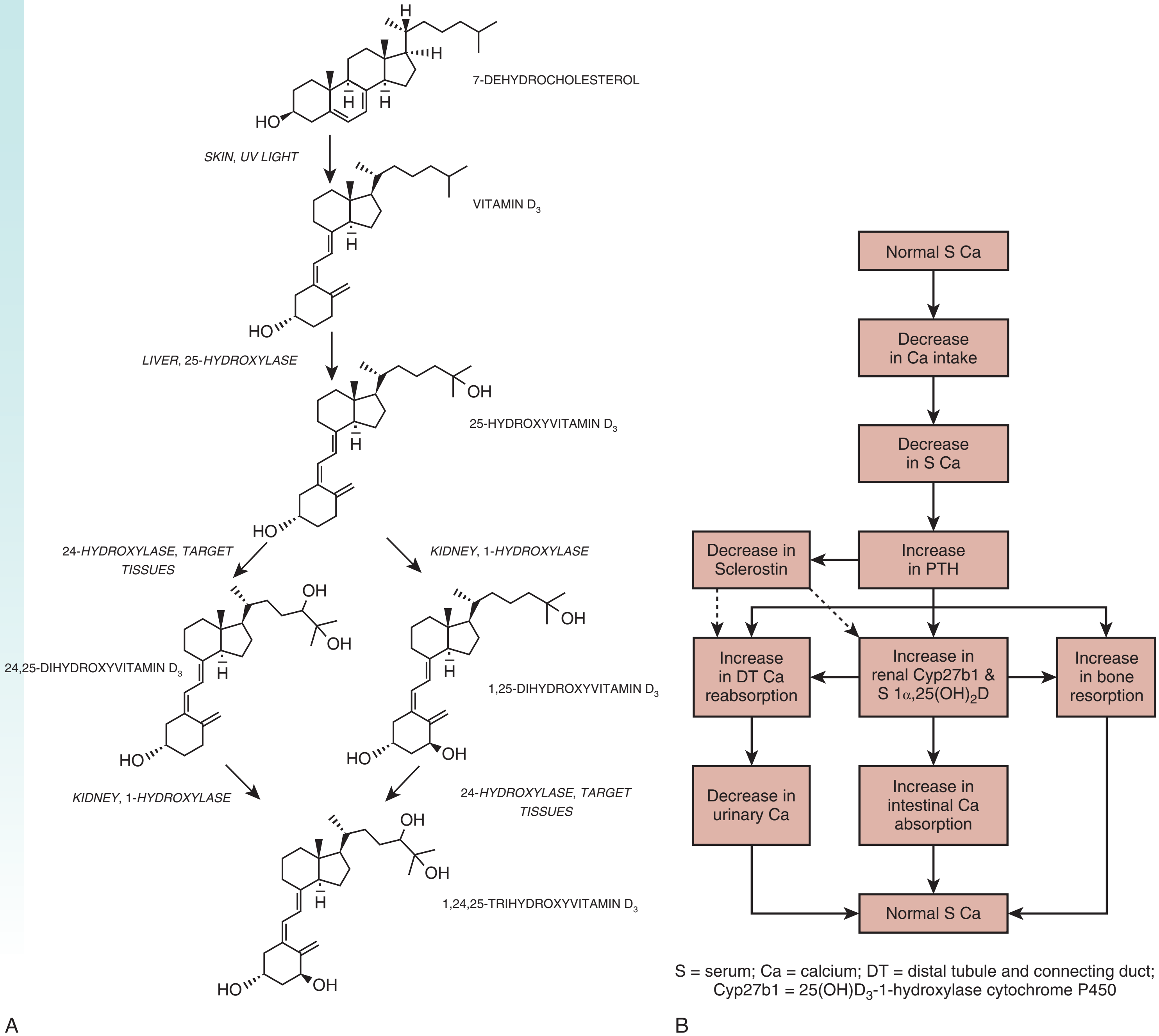


Fig. 7.4 A. The formation and metabolism of vitamin D3. B. Physiologic changes in response to decreases in serum calcium concentrations. *Ca,* Calcium; *Cyp27b1,* 25(OH)D_3-1-hydroxylase cytochrome P450; *DT,* distal tubule and connection duct; *PTH,* parathyroid hormone; *S,* serum; *UV,* ultraviolet. From Tebben PJ, Singh RJ, Kumar R. Vitamin D-mediated hypercalcemia: mechanisms, diagnosis, and treatment. *Endocr Rev.* 2016;37:521–547.

the efficiency of Pi absorption in enterocytes and proximal tubule (PT) cells.[87,106–108] In hyperphosphatemic states, renal 25-hydroxyvitamin D-1α-hydroxylase activity and 1α,25$(OH)_2D_3$ synthesis are diminished, and 25-hydroxyvitamin D-24-hydroxylase activity is increased in association with elevations in FGF-23.[7,98] Numerous other factors other than calcium and phosphorus alter the activity of the 25(OH)D-1α-hydroxylase and the reader is referred to reviews on this matter.[47,109–113]

The bioactivity of vitamin D_3 depends on the formation of 1α,25$(OH)_2D_3$ as pharmacologic amounts of vitamin D_3 or 25(OH)D_3 are required to elicit a biological response in anephric animals and patients,[53,75,114] whereas 1α,25$(OH)_2D_3$ readily increases intestinal calcium transport[49,50] and mobilizes calcium from bone[75] when given in physiologic amounts. The actions of 1α,25$(OH)_2D_3$ require the presence of the vitamin D receptor (VDR), a steroid hormone receptor, that binds 1α,25$(OH)_2D_3$ with high affinity.[115–118] Following binding of 1α,25$(OH)_2D_3$ to the ligand binding domain of the VDR, a conformational change in the receptor occurs and is associated with the recruitment of RXRα and coactivator (or corepressor) proteins to DNA binding elements within the transcription start site or other areas of genes regulated by

$1\alpha,25(OH)_2D_3$.[20,119–139] The efficiency of calcium absorption increases or decreases inversely with the amount of dietary calcium, and adaptations to changes in calcium intake are dependent on $1\alpha,25(OH)_2D_3$.[140,141] Calcium is absorbed by the intestine by passive paracellular and active transcellular mechanisms.[141–143] Active calcium absorption initially involves the movement of calcium across the apical border of the intestinal cell into the cell down a concentration and electrical gradient and does not require the expenditure of energy.[17,144] The extrusion of calcium out of the intestinal cell at the basolateral membrane is against an electrical and concentration gradient and requires the energy expenditure.[17,144] Essential to the process of active calcium transport are several vitamin D–dependent proteins, including the TRPV 5/6 (transient receptor potential vanilloid 5/6) epithelial calcium channels, calbindin D_{9K} and D_{28K}, and the plasma membrane calcium pump.[87] In the duodenal enterocyte, apically situated TRPV 5/6 cation channels mediate the increase in Ca uptake from the lumen into the cell[145]; intracellular Ca binding proteins such as calbindin D_{9K} and D_{28K} facilitate the movement of Ca across the cell[17,143]; and the basolateral plasma membrane Ca pump (PMCA)[2,146,147] and the Na-Ca exchanger (NCX)[148] assist in the extrusion of Ca from within the cell into the extracellular fluid (ECF). The Na gradient for the activity of the NCX is maintained by the Na-K-ATPase. Intestinal transcellular Ca transport is regulated by $1\alpha,25(OH)_2D_3$, which increases the expression of TRPV 6 channels,[149] the intracellular concentrations of calbindin D_{9K} and D_{28K},[17,150–152] and the expression of the plasma membrane pump, isoform 1.[153,154] The requirement of various intestinal Ca transporter proteins in transcellular Ca transport in vivo has been examined in knockout mice. Deletions of *TrpV6* and *calbindin D_{9K}* genes are not associated with alterations in intestinal Ca transport in vivo in the basal state and following the administration of $1\alpha,25(OH)_2D_3$,[155,156] although one report suggests that basal Ca transport on an adequate Ca diet is normal in *TrpV6* knockout mice but adaptations to a low-Ca diet are impaired.[157] We recently showed that deletion of the *Pmca1* in the intestine is associated with reduced growth and bone mineralization and a failure to upregulate calcium absorption in response to $1\alpha,25(OH)_2D_3$, thereby establishing the essential role of the pump in transcellular intestinal Ca transport.[158]

REABSORPTION OF CALCIUM ALONG THE TUBULE

The kidney reabsorbs filtered calcium in amounts that are subject to regulation by calciotropic hormones, PTH and $1\alpha,25(OH)_2D$.[16,145,159–164] Between 9000 and 10,000 mg of complexed and ionized calcium are filtered by the glomerulus in a 24-hour period. The amount of calcium appearing in the urine is approximately 250 mg/day, and it is therefore evident that a large percentage of filtered calcium is reabsorbed. As a result of reabsorption processes that occur in both the proximal and distal tubule, only 1%–2% of calcium filtered at the glomerulus appears in the urine.[145,161,164] Fig. 7.5 shows the percentages of calcium reabsorbed along different segments of the nephron.

CA²⁺ REABSORPTION IN THE PROXIMAL TUBULE

As noted earlier, about 60%–70% of total plasma calcium is free (not protein bound) and is filtered at the glomerulus.[165,166]

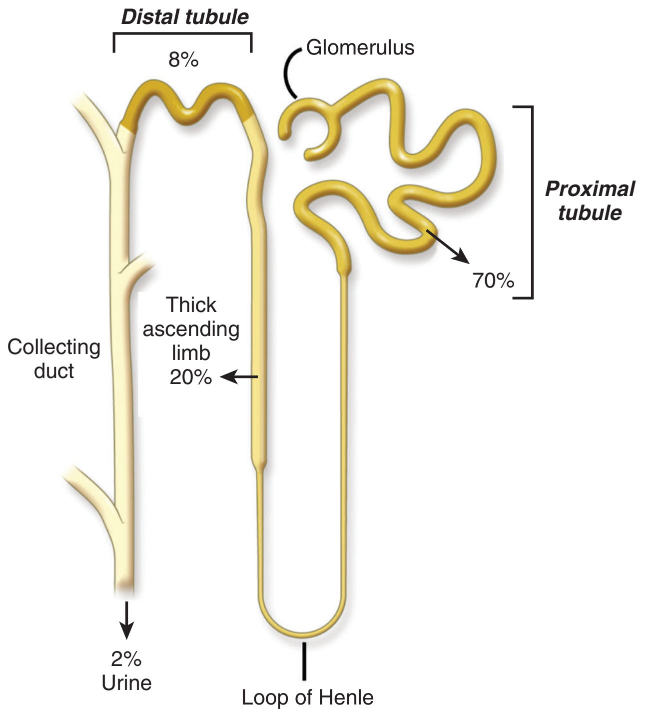


Fig. 7.5 Percentages of filtered calcium reabsorbed along the tubule.

A large percentage (~70%) of filtered calcium (Ca^{2+}) is reabsorbed in the PT mainly by paracellular processes that are linked with sodium (Na^+) reabsorption.[165,167–170] In this nephron segment, the reabsorption of Na^+ and Ca^{2+} is proportional under a variety of conditions[169,171] and is not dissociated following the administration of several factors that are known to alter renal Ca^{2+} reabsorption, such as PTH, cyclic AMP, chlorothiazide, furosemide, acetazolamide, or changes in the hydrogen ion content.[168,169,172,173] The precise cellular and molecular machinery responsible for the movement of Ca^{2+} from the lumen of the PT into the interstitial space is not clearly defined. A majority of Ca^{2+} is believed to move in between cells (paracellular movement) with a smaller, but significant, transcellular component (Fig. 7.6). The components of the paracellular pathway likely include the tight junction protein, claudin-2, which functions as a paracellular cation channel. Ca^{2+} permeates through claudin-2[174] and simultaneously competitively inhibits Na^+ conductance.[175] A transcellular component of Ca^{2+} reabsorption may also be present in the PT. Undefined Ca^{2+} channels and intracellular Ca^{2+}-binding proteins influence the movement of Ca^{2+} into and across the cell. The Na-K-ATPase has been implicated in transcellular Ca^{2+} transport in the PT,[176] and both the Na^+-Ca^{2+} exchanger[177] and isoforms 1 and 4 of the plasma membrane Ca^{2+} pump[178,179] are expressed in the PT, and could be important in the movement of Ca^{2+} out of the PT cell. Although the PT reabsorbs large amounts of Ca^{2+}, primarily by paracellular processes, the rate of Ca^{2+} reabsorption is not influenced by factors or hormones that regulate calcium balance.[168,169,172] Extracellular volume status is the major factor that influences Ca^{2+} reabsorption in the PT, via its effects on Na^+ reabsorption (see later).

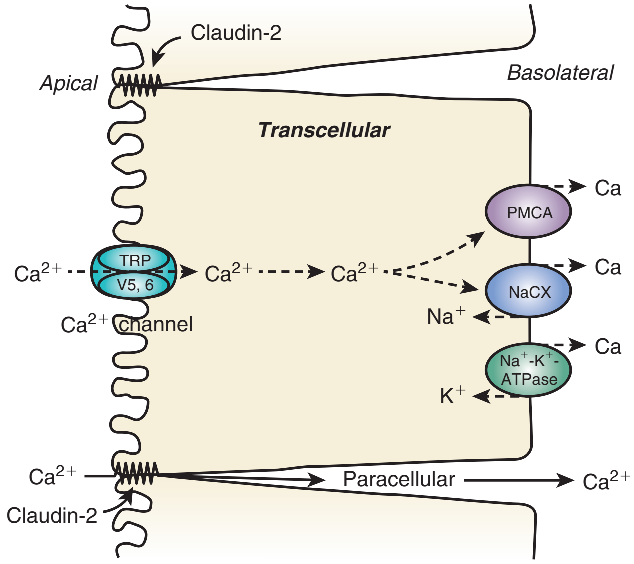


Fig. 7.6 Mechanisms by which calcium is transported in the proximal tubule. The majority of calcium is reabsorbed by paracellular mechanisms. A smaller percentage is reabsorbed by transcellular mechanisms.

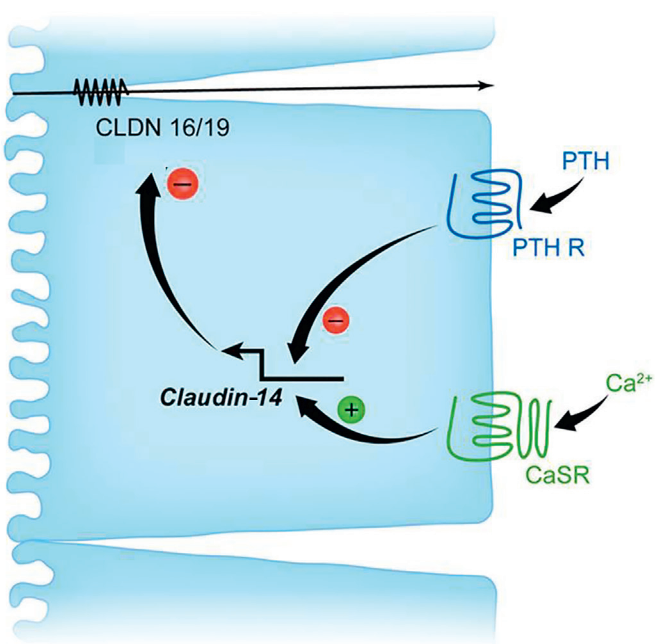


Fig. 7.7 Mechanisms and regulation of calcium transport in the thick ascending limb of Henle *(TALH)*. In the TALH, calcium is reabsorbed via paracellular mechanisms through a pore composed in part by claudin-16 and claudin-19. The activity of the latter proteins is suppressed by claudin-14, whose expression is controlled by parathyroid hormone *(PTH)* and Ca^{2+}. PTH binds to its receptor and inhibits the expression of claudin-14, whereas extracellular fluid Ca^{2+} activates the calcium-sensing receptor (CaSR) and increases claudin-14 expression. The reabsorption of Ca^{2+} in the TALH is the result of opposing activities of PTH and Ca^{2+} through their receptors. *CLDN 16/19*, Claudin-16 and -19; *PTH,* parathyroid hormone; *PTH R,* parathyroid receptor.

CA^{2+} REABSORPTION IN THE LOOP OF HENLE

The thin descending and thin ascending limbs of the loop of Henle do not transport significant amounts of $Ca.^{2+}$ [197,198] Between 20%–25% of filtered Ca^{2+} is reabsorbed in the thick ascending loop of Henle, primarily by the paracellular route involving claudin-16 and -19.[165,180–191] Thick ascending limb cells express the furosemide-sensitive Na-K-Cl cotransporter, NKCC2,[192–195] which mediates the reabsorption of Na^+ and thereby contributes to the driving force for paracellular Ca^{2+} transport. A lumen-positive transepithelial potential is generated in the thick ascending limb of the loop of Henle (TALH) through the activity of the NKCC2 (Na-K-2Cl cotransporter)[196] by two mechanisms: secondary apical recycling of K^+ via ROMK, and a NaCl diffusion potential generated by reabsorbed NaCl establishing a concentration gradient across the Na-selective paracellular pathway. This transepithelial voltage provides the driving force for passive Ca^{2+} reabsorption through the paracellular pathway.

Claudins, located in the tight junction between cells of the TALH play a role in the paracellular movement of Ca^{2+} (and Mg^{2+} reabsorption, as discussed in the next section)[197,198] (Fig. 7.7). Claudin-16 (also known as paracellin), together with claudin-19, forms a paracellular pore. A heteromeric claudin-16 and claudin-19 interaction is required to assemble and traffic to the tight junction and to generate cation-selective paracellular channels.[199,200] It has been postulated that these channels are themselves responsible for permeating divalent cations, Ca^{2+} and Mg^{2+}, via the paracellular route.[197,198] An alternative hypothesis is that claudin-16 and -19 form Na^+ channels and act primarily to establish the transepithelial NaCl diffusion potential, thus contributing to the driving force for divalent cation reabsorption.[199–202] Regardless of the mechanism, loss-of-function mutations in the genes encoding claudin-16 and -19 result in familial hypomagnesemia with hypercalciuria and nephrocalcinosis (FHHNC), which is characterized by renal Ca^{2+} and Mg^{2+} wasting due to defective thick ascending limb divalent cation reabsorption. Hou et al. generated claudin-16 and claudin-19 knockdown mouse models using transgenic siRNAs and demonstrated that these mice have significantly reduced plasma Mg^{2+} concentrations and excessive urinary excretion of Mg^{2+} and $Ca.^{2+}$ [380] Calcium deposits were observed in the basement membranes of the medullary tubules and the interstitium in the kidney of claudin-16 knockdown mice. Thus, the phenotypes of *Cldn16* and *Cldn19* knockdown mice recapitulate the phenotype of human FHHNC patients. Similarly, mutations of NKCC2 are associated with the common form of Bartter syndrome and can be associated with hypercalciuria[203] (see also Chapters 18 and 44).

Calcium-regulating hormones can regulate reabsorption of Ca^{2+} by the thick ascending limb, but there is considerable species heterogeneity. In the mouse, PTH and calcitonin (CT) stimulate Ca^{2+} transport in the cortical thick ascending limb,[184,186,204,205] whereas in the rabbit CT stimulates calcium reabsorption in the medullary thick ascending limb but not in the cortical thick ascending limb.[189] Extracellular fluid calcium also regulates calcium reabsorption in this segment through the Ca^{2+}-sensing receptor (see later).

CA^{2+} REABSORPTION IN THE DISTAL TUBULE

In the distal convoluted tubule (primarily DCT2) and connecting tubule (together abbreviated as DT), 5%–10% of filtered Ca^{2+} is reabsorbed[206–208] by active transport processes against both an electrical and concentration gradient. Ca^{2+} reabsorption in the DT occurs via a transcellular pathway. The apically situated, transient receptor potential cation channels, subfamily V, type 5 and 6 channels (TRPV5, TRPV6) mediate the increase in Ca^{2+} uptake from the lumen into the cell.[164,209–214] Micropuncture studies in knockout mice indicate that TRPV5 is the gatekeeper of Ca^{2+} reabsorption in the accessible DT in mice (Fig. 7.8A).[211] The intracellular Ca^{2+}-binding proteins, calbindin D_{9K} and D_{28K}, facilitate the movement of Ca^{2+} across the cell.[160,215] The basolateral plasma membrane calcium ATPase (PMCA) pump,[159,160,162] Na^+-Ca^{2+} exchanger (NaCX),[216–219] and the Na^+-Ca^{2+}-K^+ exchanger (NaCKX)[220] mediate the active extrusion of Ca^{2+} across the basolateral membrane (see Fig. 7.8B and 7.8C). The Na^+ gradient for the activity of the NaCX and the NaCKX is provided by the Na-K ATPase situated at the basolateral cell membrane (not shown). Ca^{2+} reabsorption in the DT is increased by PTH,[61–63] CT,[204,205] and $1\alpha,25(OH)_2D_3$.[67–71]

REGULATION OF CA^{2+} TRANSPORT IN THE KIDNEY

CALCIUM-REGULATING HORMONES

The calcium-regulating hormones, PTH and $1\alpha,25(OH)_2D_3$, regulate the expression or activity of calcium channels, calcium-binding proteins, calcium pumps, and exchangers in the kidney to increase tubule retention of filtered calcium via the transcellular pathway. PTH increases the activity of TRPV5 channels in the DT by activating cAMP-PKA signaling, and phosphorylating a threonine residue within the channel, resulting in an increase in the open probability of the channel.[220] PTH also activates the PKC pathway, thereby increasing the numbers of TRPV5 channels on the surface of tubular cells by inhibiting endocytosis of caveolae in which the channels are located.[221] $1\alpha,25(OH)_2D_3$ enhances the expression of TRPV5 and TRPV6 channels present in the distal and connecting tubule and cortical collecting duct by increasing respective mRNA concentrations through increased binding of the VDR to response elements in the gene promoters.[70,213] $1\alpha,25(OH)_2D_3$ increases the expression of calbindin D_{9K} and D_{28K} and the PMCA pump in the kidney and cultured renal cells.[70,179,222–231] The effect of PTH and $1\alpha,25(OH)_2D_3$ is to increase the expression of Ca^{2+} channels, binding proteins, pumps and exchangers, thereby increasing the retention of calcium by the kidney.

PTH has also been well described to increase renal tubule Ca^{2+} reabsorption by increasing its passive permeability in the TALH, predominantly in the cortical segment.[232,233] The mechanism has now been elucidated. PTH, acting through the PTH/PTHrP receptor, has been shown to inhibit the transcription and subcellular trafficking of claudin 14, thereby increasing paracellular Ca^{2+} reabsorption in the TALH[234] (see Fig. 7.7). Claudin-14 binds to claudin-16 and functions to inhibit the paracellular pore comprised of claudins 16 and 19. Tissue-specific knockout of the renal tubule PTH/PTHrP receptor in mice caused hypercalciuria and hypocalcemia, which was completely rescued by claudin-14 deletion, suggesting that the effect of PTH on paracellular Ca^{2+} transport in the TALH may be more important than previously appreciated.

EXTRACELLULAR CALCIUM AND DIET

The level of extracellular calcium regulates renal Ca^{2+} reabsorption by signaling though the Ca-sensing receptor (CaSR). In the kidney, the CaSR is primarily expressed on the basolateral membrane of the TALH. Activation of CaSR reduces renal tubular Ca^{2+} reabsorption and induces calciuresis

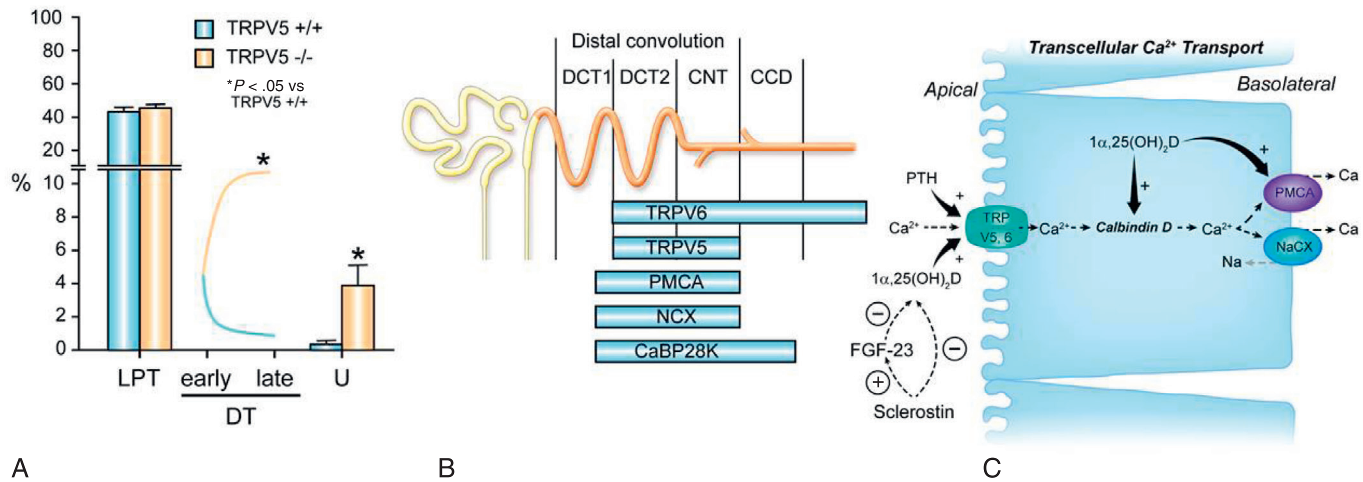


Fig. 7.8 Mechanisms of calcium transport in the distal convoluted tubule. A. Role of TRPV5 investigated by micropuncture of kidneys from TRPV5 knockout mice. The figure shows fractional Ca^{2+} delivery to micropuncture sites in the late proximal tubule *(LPT)* to sites located along the distal convolution *(DC)* from early to late DC (as localized using tubule K^+ concentrations) and to the urine. Deletion of TRPV5 in mice prevents Ca^{2+} reabsorption along the DT and there is even evidence for Ca^{2+} leaking back into the lumen, possibly by paracellular routes; TRPV6 may partially compensate in the collecting duct. **B.** Distribution of $1\alpha,25(OH)_2D$ or parathyroid hormone–sensitive channels and transporters along the distal convoluted tubule (*DCT1* and *DCT2*), connecting tubule *(CNT),* and cortical and medullary collecting ducts (*CCD* and *MCD*). **C.** Ca transport in the DT occurs by transcellular mechanisms. Transcellular Ca transport is mediated by several channels, pumps, and exchangers located at the apical and basolateral portions of the cell. Modified from Kumar R, Vallon V. Reduced renal calcium excretion in the absence of sclerostin expression: evidence for a novel calcium-regulating bone kidney axis. *J Am Soc Nephrol.* 2014;25:2159–2168, with permission of the publisher.

in response to a Ca load.[235,236] One mechanism is by inhibition of NKCC2 expression[236] or activity. More recently, it has been suggested that CaSR acts primarily by regulating paracellular permeability. Loupy et al. showed that a CaSR antagonist increased Ca^{2+} permeability in isolated perfused TALH with no change in transepithelial voltage or Na flux.[237] This appears to be mediated by regulation of the expression of claudin-14. Activation of the CaSR causes robust upregulation of claudin-14,[238,239] which through physical interaction, inhibits paracellular cation channels formed by claudin-16 and claudin-19[238] (see Fig. 7.7). The signaling mechanism seems to involve CaSR inhibiting calcineurin, a phosphatase that normally activates NFAT to increase transcription of two micro-RNAs miR-9 and miR-374, thereby downregulating claudin-14 expression.[240,241] The central role of claudin-14 is further supported by the finding that claudin-14 knockout mice are unable to increase their fractional excretion of calcium in response to a high Ca^{2+} diet,[238] and exhibit complete loss of regulation of urinary Ca^{2+} excretion in response to either a CaSR agonist or antagonist.[240,242] Of note, claudin-14 expression is also decreased in kidney of mice fed a low-calcium diet compared with those fed a high-calcium diet,[234] providing a mechanism by which calcium excretion is matched to dietary intake.

DIURETICS

Loop diuretics such as furosemide increase urinary calcium losses. The mechanism by which furosemide causes hypercalciuria is linked to its ability to bind to and inhibit the furosemide-sensitive Na-K-Cl cotransporter, NKCC2,[192–195] present in the TALH. NaCl absorption is diminished, as is potassium recycling, resulting in a reduction in lumen positivity that drives Ca^{2+} reabsorption. Subjects with the common form of Bartter syndrome have inactivating mutations of the NKCC2 and associated with calciuria.[203] Compensatory increases occur in the expression of distal tubule transport channels and proteins such as the TRPV5 and TRPV6 channels and calbindin D_{28K} following the administration of furosemide, but fail to compensate for the increase in excretion that occurs in the TALH.[242] Thiazide diuretics, on the other hand, cause hypocalciuria[213,243–245] and the effect appears to be independent of PTH in humans and rodents. Thiazides bind to and inhibit the Na-Cl cotransporter in the distal tubule.[192,246] Chronic thiazide use is associated with a reduction in extracellular fluid volume, which secondarily enhances Na^+ and Ca^{2+} reabsorption in the PT of in the kidney.[173] Distal tubule Ca^{2+} transport is clearly unaffected by "chronic" thiazide use,[173] in contrast to older reports that thiazide "acutely" increases Ca^{2+} reabsorption in isolated perfused DCT.[247] The development of hypocalciuria parallels a compensatory increase in Na^+ reabsorption secondary to an initial natriuresis following thiazide administration. These observations are supported by the upregulation of the Na^+/H^+ exchanger, responsible for the majority of Na^+ and associated Ca^{2+} reabsorption in the PT, whereas the expression of proteins involved in active Ca^{2+} transport in the distal tubule was unaltered. Indeed, thiazide administration was associated with hypocalciuria in *Trpv5*-knockout mice. Humans with Gitelman syndrome and inactivating mutations of the thiazide-sensitive Na-Cl transporter have hypocalciuria, hypomagnesemia, and volume depletion,[208,248–250] findings that are recapitulated in the Na-Cl cotransporter knockout mouse.[251]

Clinical Relevance

Hypomagnesemia With Epidermal Growth Factor (EGF) Receptor Inhibitors

Because EGF activates TRPM6 in the distal convoluted tubules, EGF receptor inhibitors, which are widely used as chemotherapeutic agents, cause hypomagnesemia. Monoclonal antibodies targeting the EGF receptor, such as cetuximab and panitumumab, are frequently associated with this complication, whereas the incidence with small-molecule tyrosine kinase inhibitors such as erlotinib seems to be less.

Diuretics in Calcium Disorders

Because of their hypercalciuric effect, loop diuretics are used for the acute treatment of hyperkalemia. Conversely, in patients with idiopathic hypercalciuria, thiazide diuretics are effective at lowering urine calcium concentration and thereby reducing the risk of kidney stone formation.

ESTROGENS

Estrogens influence calcium transport in the kidney as postmenopausal women have higher urinary Ca^{2+} excretion than premenopausal women.[252] In the early postmenopausal period the administration of estrogen is associated with a decrease in urine Ca^{2+} excretion and an increase in serum PTH and $1\alpha,25(OH)_2D$.[253,254] Estradiol increases the expression of the TRPV5 channel in the kidney in a manner independent of $1\alpha,25(OH)_2D_3$.[255] These observations are supported by reduced duodenal TRPV5 channel expression in mice lacking the estrogen receptor α.[256]

EXTRACELLULAR FLUID VOLUME

In conditions such as volume depletion, where PT Na^+ reabsorption is increased, one also observes enhanced Ca^{2+} reabsorption that can contribute to the hypercalcemia that is sometimes seen in such situations. Conversely, the salutary effects of isotonic saline administration in hypercalcemic patients are attributable to a reduction in Ca^{2+} reabsorption as a result of reduced Na^+ reabsorption.

METABOLIC ACIDOSIS AND ALKALOSIS

Metabolic acidosis is associated with hypercalciuria, and when prolonged, often results in bone loss and osteoporosis.[257] Metabolic acidosis and metabolic alkalosis decrease or increase the reabsorption of Ca^{2+} in the distal tubule,[172,258–261] the expression TrpV5 in the distal tubule,[262] and the activity of TrpV5 channels.[263–265]

REGULATION OF RENAL CALCIUM TRANSPORT BY NOVEL PROTEINS

KLOTHO

Klotho is a coreceptor for the phosphaturic peptide, FGF 23, with β-glucuronidase activity.[266–269] It is a kidney- and parathyroid gland–specific protein, which influences epithelial Ca^{2+} transport by deglycosylating TRPV5, thereby trapping the channel in the plasma membrane and sustaining the activity of the channel.[270] Further evaluation of serum Klotho concentrations and their association with changes in renal calcium excretion is required to establish a role of this factor in regulation of renal calcium transport.

SCLEROSTIN

Sclerostin is an osteocyte-derived glycoprotein that influences bone mass.[271] Patients with sclerosteosis and its milder variant, van Buchem disease,[272–274] have exceptionally dense bones and skeletal overgrowth that often constricts cranial nerve foramina and the foramen magnum, resulting in premature death. Sclerosteosis is due to inactivating mutations of the sclerostin (*SOST*) gene, and the milder van Buchem disease is due to a 52-kb deletion of a downstream enhancer element of the sclerostin gene.[275] Mouse models of sclerosteosis have increases in skeletal mass similar to those found in patients with the disease.[55,276–278] By using a *Sost* gene knockout model generated in our laboratory[55] we have demonstrated that sclerostin, either directly or indirectly, through an alteration in the synthesis of 1α,25-dihydroxyvitamin D (1α,25$(OH)_2$D), influences renal calcium reabsorption in the kidney. Urinary calcium excretion and renal fractional excretion of calcium are decreased in $Sost^{-/-}$ mice.[55] Serum 1α,25$(OH)_2$D concentrations are increased without attendant hypercalcemia; renal 25(OH)D-1α hydroxylase (*Cyp27b1*) mRNA and protein expression are also increased in $Sost^{-/-}$ mice, strongly suggesting that the increase in serum 1α,25$(OH)_2$D concentrations was due to increased 1α,25$(OH)_2$D synthesis. When recombinant sclerostin is added to cultures of proximal tubular cells the expression of the messenger RNA for *Cyp27b1,* the 1α-hydroxylase cytochrome P450, is diminished. Serum 24,25$(OH)_2$D concentrations were diminished in $Sost^{-/-}$ mice, and PTH concentrations were similar in knockout and wild-type mice. The lack of change in PTH is consistent with previous studies in humans.[279] The data suggest that in addition to the hormones traditionally thought to alter calcium reabsorption in the kidney (PTH and 1α,25$(OH)_2$D), sclerostin plays a significant role in altering renal calcium excretion. Wheras PTH and 1α,25$(OH)_2$D decrease fractional excretion of calcium by increasing the efficiency of calcium reabsorption in the DT, sclerostin increases fractional excretion of calcium (the absence of sclerostin expression being associated with a reduced fractional excretion of calcium).[55] Thus, the adaptation to a reduction in calcium intake and resultant downstream alterations in hormones may need to be amended to include changes in sclerostin expression (see Fig. 7.4B and Fig.7.8C).

MAGNESIUM TRANSPORT IN THE KIDNEY

THE ROLE OF MAGNESIUM IN CELLULAR PROCESSES

Magnesium is an abundant cation in the human body (Table 7.1).[280–285] Magnesium is required for a variety of biochemical functions.[286] The activities of magnesium-dependent enzymes are modulated by the metal as a result of binding to the substrate or as a result of direct binding to the enzyme.[286–289] Enzymes of the glycolytic and citric acid pathways, exonuclease, topoisomerase, RNA and DNA polymerases, and adenylate cyclase are among the many enzymes regulated by magnesium.[286–290] Additionally, magnesium regulates channel activity.[286]

Table 7.2 Distribution and Concentrations of Magnesium in a Healthy Adult

Site	% total-body Mg	Concentration/content
Bone	53	0.5% of bone ash
Muscle	27	9 mmol/kg wet weight
Soft tissue	19	9 mmol/kg wet weight
Adipose tissue	0.012	0.8 mmol/kg wet weight
Erythrocytes	0.5	1.65–2.73 mmol/L
Serum	0.3	0.69–0.94 mmol/L

Given its role in such diverse biological processes, it is not surprising that a deficiency or increase in serum magnesium concentrations is associated with important clinical symptoms.[291] For example, low magnesium concentrations are associated with muscular weakness, fasciculations, Chvostek and Trousseau signs, and sometimes frank tetany.[291] The tetany of hypomagnesemia is independent of changes in serum calcium. On occasion, personality changes, anxiety, delirium and psychoses may manifest. Hypocalcemia,[292–298] reduced PTH secretion,[299–304] and hypokalemia[305–308] are sometimes present in hypomagnesemic subjects. Cardiac arrhythmias and prolongation of the corrected QT interval[309,310] are sometimes observed. Conversely, hypermagnesemia seen in association with the administration of excessive amounts of magnesium in diseases such as eclampsia and in patients with renal failure is manifest as weakness of the voluntary muscles.

MAGNESIUM IS PRESENT IN SERUM IN BOUND AND FREE FORMS

Most magnesium within the body is present in bone or within the cells (Table 7.2).[286] Approximately 60% of magnesium is stored in bone. Serum magnesium concentrations vary slightly with age and in adults are 1.6–2.3 mg/dL (0.66–0.94 mmol/L). Within plasma, about 70% of Mg is ultrafiltrable, 55% is free, and about 14% of Mg is in the form of soluble complexes with citrate and phosphate.[311] Because Mg is present largely within cells and bone, there is some interest as to whether serum Mg concentrations reflect tissue stores, especially when Mg is depleted or deficient. When rats[312–314] and humans[291,315] are fed Mg-deficient diets, serum Mg decreases within 1 day in rats and in 5–6 days in humans. Bone Mg and blood mononuclear cell Mg concentrations correlate well with total body Mg and serum Mg.[315–318] The correlations between total body Mg stores and muscle or cardiac Mg, however, are not precise.[315]

REGULATION OF MG HOMEOSTASIS

The intestine and the kidney regulate magnesium balance (Fig. 7.9).[286] A diet adequate in magnesium normally contains 200–300 mg of magnesium.[319] Between 75 and 150 mg of ingested dietary magnesium is absorbed in the jejunum and ileum, primarily by paracellular passive processes.[320–325] The TRPM6 protein (a mutant form of this protein is present in patients with familial hypomagnesemia) is localized to

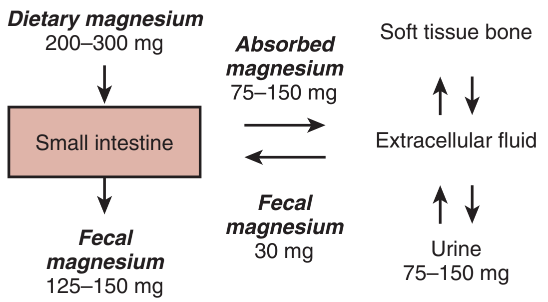


Fig. 7.9 Magnesium homeostasis in normal humans showing the amounts of magnesium absorbed in the intestine and reabsorbed by the kidney.

the apical membrane of intestinal and renal epithelial cells and mediates transcellular magnesium absorption.[145,326] A homolog of TRPM6, TRPM7 is ubiquitously expressed and plays a role in whole-body magnesium homeostasis and many cellular functions, ranging from control of cell proliferation and cellular magnesium homeostasis to cell adhesion and cell migration.[327–335] It forms a heteromeric complex with TRPM6 and is necessary for TRPM6 activity and epithelial magnesium absorption.[336,337] About 30 mg of magnesium is secreted into the intestine via pancreatic and intestinal secretions, giving a net magnesium absorption of approximately 130 mg/24 h. Magnesium that is not absorbed in the intestine and is secreted into the intestinal lumen eventually appears in the feces (125–150 mg). Absorbed magnesium enters the extracellular fluid pool and moves in and out of bone and soft tissues. Approximately 130 mg of magnesium (equivalent to the net and amount absorbed in the intestine) is excreted in the urine.

In experimental animals and humans, feeding a diet low in magnesium results in a rapid decrease in urinary and fecal magnesium and the occurrence of a negative magnesium balance.[338–347] Conversely, the administration of magnesium is associated with an increase in the renal excretion of magnesium.[348–350] Unlike the cases of calcium and phosphorus, however, no hormones or molecules have been identified that alter magnesium transport in the intestine or that alter the renal excretion of magnesium in response to changes in magnesium balance.[286,351,352]

REABSORPTION OF MAGNESIUM ALONG THE TUBULE

Approximately 10% of total body magnesium is filtered daily by the glomeruli (approximately 3000 mg/24 hr). About 75% of total plasma magnesium is filterable. Because urinary magnesium excretion is about 150 mg/24 hr, a substantial fraction of filtered magnesium is reabsorbed along the tubule (approximately 95%).

MG^{2+} REABSORPTION IN THE PROXIMAL TUBULE

Between 15% and 20% of filtered magnesium is reabsorbed in the PT (Fig. 7.10). The cellular and molecular mechanisms by which magnesium is reabsorbed in the proximal nephron are unknown. However, it is speculated that reabsorption of magnesium in the proximal nephron occurs by paracellular mechanisms, likely driven by the concentration gradient

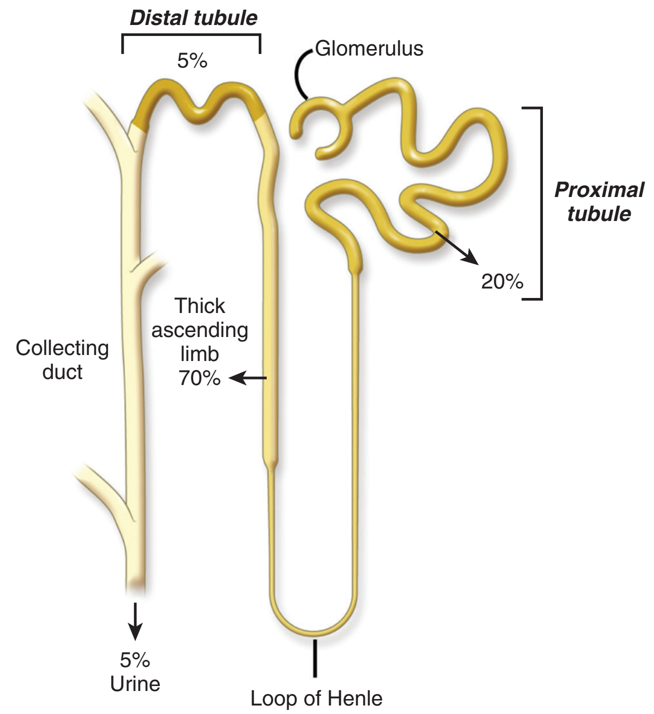


Fig. 7.10 Percentages of filtered magnesium reabsorbed along the tubule.

resulting from reabsorption of Na^+ and water. However, magnesium permeability in this segment is likely to be quite low, as the tubular fluid-to-ultrafiltrate magnesium ratio can rise to 1.65 in the late PT.[353]

MG^{2+} REABSORPTION IN THE THICK ASCENDING LIMB

The bulk of the filtered magnesium is reabsorbed in the TALH, again by a paracellular mechanism of which claudin-16 is a critical component.[199,201,354–361] As discussed earlier for Ca^{2+} transport, claudin-16 and claudin-19 form cation-selective paracellular channels[199,200] that either directly mediate paracellular Mg^{2+} reabsorption or facilitate the generation of an NaCl diffusion potential that provides the driving force for paracellular Mg^{2+} reabsorption. Mutations of the *CLDN16* and *CLDN19* genes, and the *SLC12A1, KCNJ1,* and *CLCNKB* genes that encode proteins required for normal thick ascending limb function, result in excessive magnesium losses in the urine and hypomagnesemia. Claudin-10 is also highly expressed in the TALH. The predominant isoform, claudin-10b, acts as a paracellular Na^+ channel and is spatially distinct from claudin-16/19, being expressed mainly in the medullary TALH.[362] Deletion of the *Cldn10* gene in mice is associated with decreased paracellular sodium reabsorption, hypermagnesemia, and nephrocalcinosis.[363] In isolated perfused TAL tubules of claudin-10–deficient mice, paracellular permeability of sodium is decreased and the relative permeability of calcium and magnesium is increased. This suggests that claudin-10b uses the lumen-positive voltage in the early TALH to drive paracellular Na^+ reabsorption and thereby reduces the electrical potential available for divalent cation reabsorption later on in the cortical TALH. Mutations in claudin-10, or specifically in claudin-10b, have recently been described in several families with variable degrees of hypermagnesemia, hypocalciuria, and hypokalemic metabolic alkalosis, together with several unusual extrarenal manifestations, including anhidrosis, alacrima, xerostomia, and ichthyosis.[364–366] Fig. 7.11 shows the mechanism by which magnesium is transported in the TALH.

Magnesium reabsorption in the TALH is inhibited by hypermagnesemia, presumably because it reduces the concentration gradient for paracellular diffusion.[367] Conversely, hypomagnesemia and magnesium depletion stimulate magnesium reabsorption in the TALH.[368] These are also the main physiological regulators of renal magnesium excretion.

Mg^{2+} REABSORPTION IN THE DISTAL TUBULE

Between 5% and 10% of filtered magnesium is reabsorbed transcellularly in the distal convoluted tubule. The rate-limiting step is thought to be apical entry of Mg^{2+} through an Mg^{2+} channel that is formed by a complex of TRPM6 and TRPM7.[326,336,337] Epidermal growth factor (EGF) promotes TRPM6 trafficking to the plasma membrane.[369] The driving force for apical Mg^{2+} entry is the lumen-negative electrical membrane potential, the set point of which is likely determined by the conductance of an apical voltage-gated potassium channel, Kv1.1.[370] These explain why mutations in TRPM6, pro-EGF, and Kv1.1 are all genetic causes of hypomagnesemia (see Chapter 44 for a detailed discussion of inherited hypomagnesemia).

It is unclear as to the mode of basolateral exit of magnesium from the cell into the interstitial space. CNNM2 is suspected to play a role because it encodes a transmembrane protein localized at the basolateral membrane that is regulated by Mg^{2+} deficiency and when mutated causes renal Mg^{2+} wasting.[371] It has been proposed to function as an Mg^{2+} channel or Mg^{2+}-sensitive Na^+ channel,[372,373] but others have not found evidence that it transports Mg^{2+}.[374]

Another intriguing possibility is SLC41A1, a homolog of bacterial MgtE Mg^{2+} transporters[375] that functions as an Na^+-Mg^{2+} exchanger when expressed in mammalian cells.[376] Interestingly, mutations in SLC41A1 have recently been found to cause a form of nephronopthisis.[377] Because basolateral extrusion of Mg^{2+} in the DT must involve energetically active transport, it is likely that the Na-K-ATPase plays a role, albeit indirectly. FXYD2 likely participates in this because it is the gamma subunit of the Na-K-ATPase and, when mutated, causes dominant isolated hypomagnesemia. Likewise, the transcription cofactors, hepatocyte nuclear factor 1B (HNF1B) and pterin-4 alpha-carbinolamine dehydratase (PCBD1) costimulate the promoter of FXYD2, and so mutations in either of them are associated with hypomagnesemia.[378,379] Finally, mutations in the basolateral potassium channel in the DT, Kir4.1, cause a syndrome called SeSAME or EAST that is associated with hypomagnesemia.[380,381] Kir4.1 is thought to facilitate Mg^{2+} reabsorption in the DT by recycling K^+ across the basolateral membrane, thus enabling the Na-K-ATPase to transport Na^+. Fig. 7.12 shows the cellular localization of these proteins in the distal convoluted tubule into the cell.

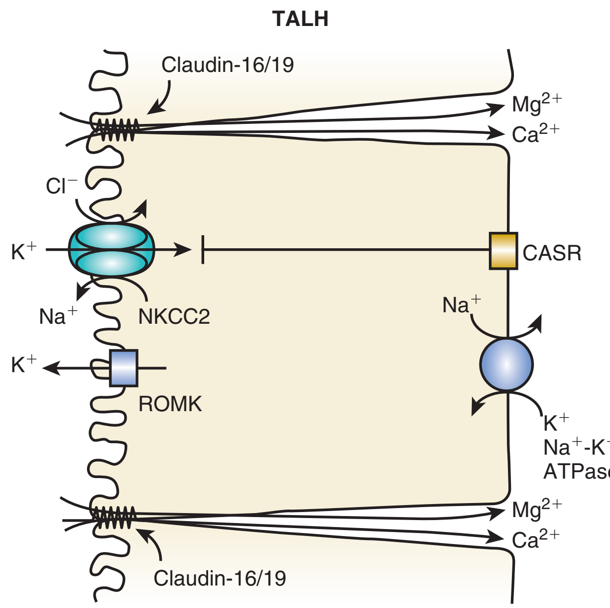


Fig. 7.11 Mechanism by which magnesium is transported in cells of the thick ascending limb of the loop of Henle *(TALH)*. The majority of magnesium is reabsorbed by paracellular mechanisms. Claudin-16 and claudin-19 are depicted as directly transporting Ca^{2+} and Mg^{2+}, but it has also been hypothesized that their primary role is to allow backleak of reabsorbed Na^+ and thus establish an NaCl diffusion potential, thereby indirectly facilitating divalent cation reabsorption. *CaSR,* Calcium-sensing receptor; *NKCC2,* Na-K-Cl cotransporter; *ROMK,* renal outer medullary potassium channel.

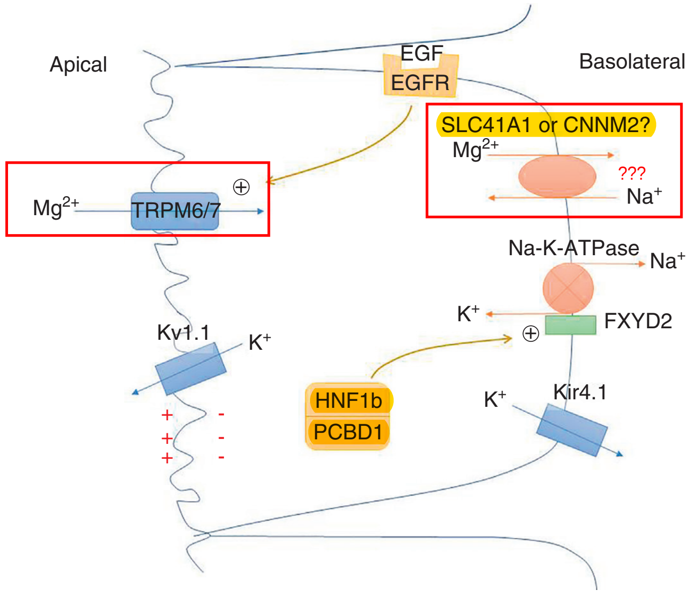


Fig. 7.12 Mechanism by which magnesium is transported in distal tubule cells. The majority of magnesium is reabsorbed by transcellular mechanisms. Mg^{2+} enters apically via the TRPM6/7 heteromeric channel, driven by the membrane voltage set by the K^+ channel, Kv1.1. Its exit pathway is unknown but may be via CNNM2 or SLC41A1, perhaps acting as a basolateral Na–Mg exchanger. Active extrusion is likely driven by the Na^+ gradient generated by the Na-K-ATPase alpha subunit and accessory subunit, FXYD2. The basolateral K^+ channel Kir 4.1 recycles K^+ that enters via the Na-K-ATPase. See text for role of other regulators. *EGF,* Epidermal growth factor; *EGFR, epidermal growth factor receptor.*

REGULATION OF MAGNESIUM TRANSPORT IN THE KIDNEY

A variety of factors alter magnesium reabsorption in the kidney (Table 7.3)*. With the exception of magnesium excess and depletion, it is unclear whether any of these are physiologically important regulators. Thus, although effects on magnesium excretion in the urine are noted following the infusion of various substances or following blockade of their activity, it is not clear that such changes occur with physiologic changes in concentrations of these factors in vivo. Furthermore, the concentrations of the effector substances do not change in the physiologically appropriate manner following changes in serum concentrations of magnesium. Thus, hormonal homeostasis, in which concentrations of a hormone (PTH, glucagon, arginine vasopressin, and so on) change following changes in the concentration of magnesium, and in turn, alter the retention or concentrations of magnesium, is difficult to demonstrate.

PHOSPHORUS TRANSPORT IN THE KIDNEY

THE ROLE OF PHOSPHORUS IN CELLULAR PROCESSES

Phosphorus is a key component of hydroxyapatite, the major component of bone mineral, nucleic acids, bioactive signaling proteins, phosphorylated enzymes, and cellular membranes.[411–414] Prolonged deficiency of phosphorus and inorganic phosphate results in serious biological problems, including impaired bone mineralization, resulting in osteomalacia or rickets, abnormal erythrocyte, leukocyte and platelet function, impaired cell membrane integrity that can result in rhabdomyolysis, and impaired cardiac function.[415–417] Phosphate balance is maintained through a series of complex hormonally and locally regulated metabolic adjustments. In states of neutral phosphate balance, net intake equals net output. The major organs involved in the absorption, excretion, and reabsorption of phosphate are the intestine and the kidney (Fig. 7.13). A normal diet adequate in phosphorus normally contains ~1500 mg of phosphorus. Approximately 1100 mg of ingested dietary phosphate is absorbed in the proximal intestine predominantly in the jejunum. About 200 mg of phosphorus is secreted into the intestine via pancreatic and intestinal secretions, giving a net phosphorus absorption of approximately 900 mg/24 hr. Phosphorus that is not absorbed in the intestine or is secreted into the intestinal lumen eventually appears in the feces. Absorbed phosphorus enters the extracellular fluid pool and moves in and out of bone (and to a smaller extent in and out of soft tissues) as needed (~200 mg). Approximately 900 mg of phosphorus (equivalent to the amount absorbed in the intestine) is excreted in the urine.

Table 7.3 Factors Altering the Reabsorption of Magnesium in the Kidney

Substance	Effect
Peptide hormones	
Parathyroid hormone[205,383–387]	Increase
Calcitonin[387–395]	Increase
Glucagon[396,397]	Increase
Arginine vasopressin[398]	Increase
Insulin[399]	Increase
β-Adrenergic agonists	
Isoproterenol[382]	Increase
Prostaglandins PGE_2[400]	Decrease
Mineralocorticoids	
Aldosterone	Increase
1, 25-dihydroxyvitamin D_3[401]	Decrease
Magnesium	
Restriction[338–341,343]	Increase
Increase[348–350]	Decrease
Metabolic alkalosis[261,402,403]	Increase
Metabolic acidosis[261,402,403]	Decrease
Hypercalcemia[404]	Decrease
Phosphate depletion[405,406]	Decrease
Diuretics	
Furosemide	Decrease
Amiloride[408,409]	Increase
Chlorothiazide[407,410]	Increase

PGE_2, Prostaglandin E_2.
From Dai LJ, Ritchie G, Kerstan D, et al. Magnesium transport in the renal distal convoluted tubule. *Physiol Rev*. 2001;81:51–84.

*References 205, 261, 338–341, 343, 348–350, and 382–410.

PHOSPHORUS IS PRESENT IN BLOOD IN MULTIPLE FORMS

About 85% of phosphorus in the body is present in bones, 14% exists in cells from soft tissues, and 1% is present in extracellular fluids. In mammals, bone contains a substantial amount of phosphorus (approximately 10 g/100 g dry fat free tissue); in comparison, muscle contains 0.2 g/100 g fat free tissue, and the brain 0.33 g/100 g fresh tissue.[418] Phosphorus is present in virtually every bodily fluid. In human plasma

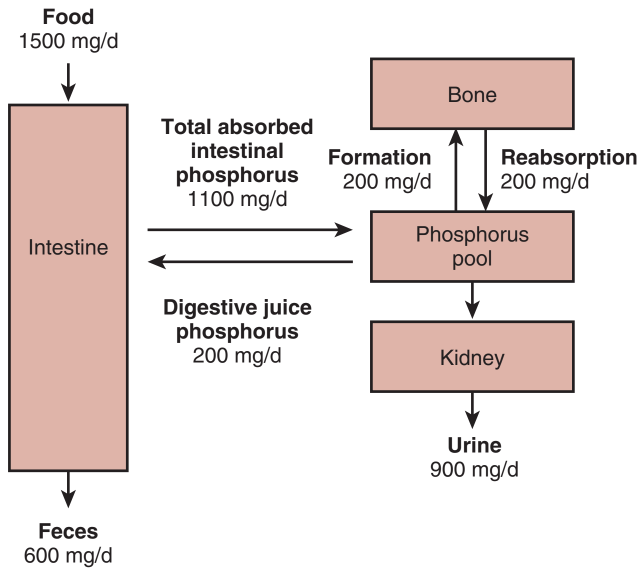


Fig. 7.13 Phosphorus homeostasis in humans. The major organs involved in the absorption, excretion, and reabsorption of phosphate are the intestine and the kidney.

Table 7.4 Distribution of and Concentrations of Phosphorus (mmol/L) in Adult Human Blood

Phosphorus Compounds	Erythrocytes	Plasma
Phosphate ester	12.3–19.0	0.86–1.45
Phospholipids	4.13–4.81	2.23–3.13
Inorganic phosphate	0.03–0.13	0.71–1.36

or serum, phosphorus exists in the form of inorganic phosphorus or phosphate (Pi), lipid phosphorus, and phosphoric ester phosphorus. Total serum phosphorus concentrations range between 8.9 and 14.9 mg/dL (2.87–4.81 mmol/L), inorganic phosphorus (phosphate, Pi) concentrations between 2.56 and 4.16 mg/dL (0.83–1.34 mmol/L) (this is what is usually measured clinically and referred to as the serum phosphorus, and the normal range changes with age),[419] phosphoric ester phosphorus concentrations between 2.5 and 4.5 mg/dL (0.81–1.45 mmol/L), and lipid phosphorus concentration between 6.9 and 9.7 mg/dL (2.23–3.13 mmol/L) (Table 7.4).[418]

REGULATION OF PHOSPHATE HOMEOSTASIS—AN INTEGRATED VIEW

Intestinal feed-forward and hormonal feedback systems (PTH–vitamin D endocrine system and the phosphatonins) are likely to be responsible for the control of phosphorus homeostasis (Fig. 7.14). The short-term responses that occur within minutes to hours of feeding of a high-Pi meal are important in regulating phosphorus homeostasis via feed-forward mechanisms, whereas the longer-term changes occur as a result of alterations in circulating concentrations of PTH, 1α,25$(OH)_2D_3$, and the phosphatonins such as fibroblast growth factor 23.[86,420–423] Intestinal signals have been shown in rodents to rapidly alter renal Pi excretion in response to changes in duodenal Pi concentrations.[421]

PTH, 1α,25$(OH)_2D_3$, and the phosphatonin FGF-23 control phosphorus homeostasis on longer-term basis (hours to days).[85,86] Concentrations of these hormones and factors are regulated by phosphorus in a manner that is conducive to the maintenance of normal phosphorus concentrations. Fig. 7.15 shows the physiologic changes known to occur with low or high dietary intakes of phosphate. A decrease in serum phosphate concentrations, as would occur with a reduced intake of phosphorus, results in increased ionized calcium concentrations, decreased PTH secretion, and a subsequent decrease in renal phosphate excretion. At the same time, by PTH-independent mechanisms, there is an increase in renal 25-hydroxyvitamin D 1α-hydroxylase activity, increased 1α,25$(OH)_2D_3$ synthesis, and increased phosphorus absorption in the intestine and reabsorption in the kidney.[88,92,94,99–105] Conversely, with elevated phosphate intake, there are decreased calcium concentrations and increased PTH release from the parathyroid gland. PTH actually has two opposing effects: it increases urinary phosphate excretion but also increases the synthesis of 1α,25$(OH)_2D_3$ by stimulating the activity of the renal 25-hydroxyvitamin D 1α-hydroxylase; the net effect is to increase renal phosphate excretion. Increased serum phosphate concentrations simultaneously inhibit renal 25-hydroxyvitamin D 1α-hydroxylase and decrease 1α,25$(OH)_2D_3$ synthesis. Reduced 1α,25$(OH)_2D_3$ concentrations decrease intestinal phosphorus absorption as well as renal

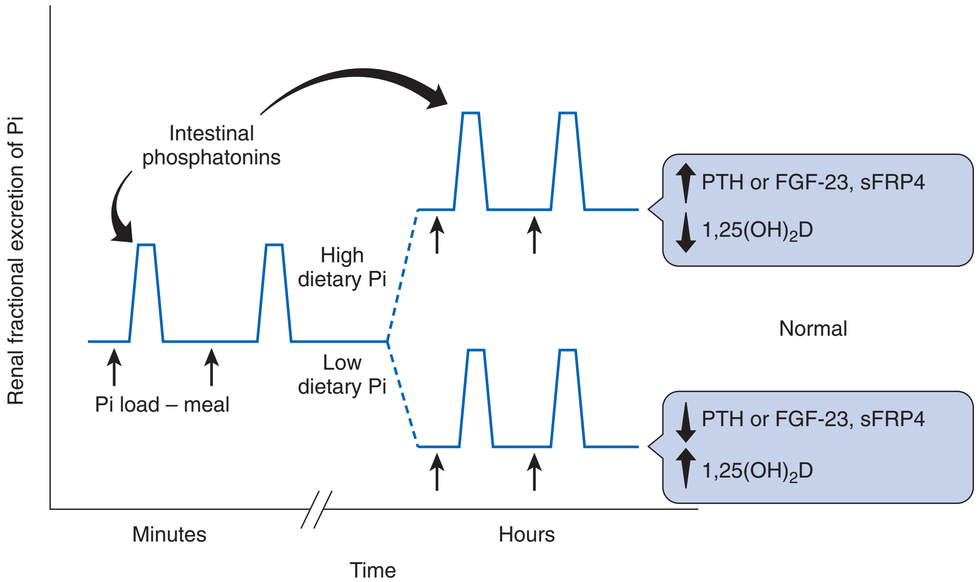


Fig. 7.14 Intestinal feed-forward and hormonal feedback systems are responsible for the control of phosphorus homeostasis. Changes in intestinal luminal phosphate concentrations *(Pi load – meal)* result in the elaboration of chemical signals *(Intestinal phosphatonins)* that alter the fractional excretion of phosphate in the kidney over a time frame of minutes. Long-term changes in the amount of phosphate in the diet result in changes in the concentrations of PTH, 1α,25-dihydroxyvitamin D, and phosphatonins, which influence the fractional excretion of phosphate in the kidney over a time frame of hours, as shown. Short-term changes mediated by feed-forward intestinal signals are proposed to be superimposed on this chronic baseline. *PTH,* Parathyroid hormone.

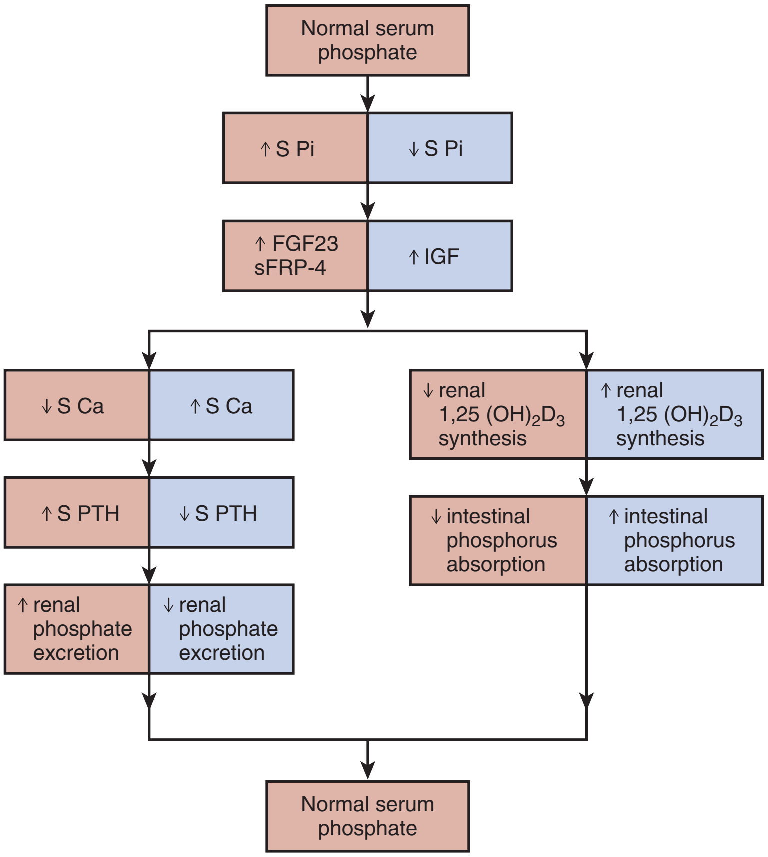


Fig. 7.15 Changes in growth factors (fibroblast growth factor 23 *[FGF 23], sFRP-4* (secreted frizzled related protein 4) insulin-like growth factor *(IGF),* parathyroid hormone *(PTH),* and 1α,25-dihydroxyvitamin D, and the subsequent physiologic changes in intestinal phosphate absorption or renal phosphate reabsorption following perturbations in serum phosphate.

phosphate reabsorption. All of these factors tend to bring serum phosphate concentrations back into the normal range.

The phosphatonins FGF-23 and sFRP-4 inhibit renal phosphate reabsorption.[98,424–432] They also decrease,[424,430,433–438] and IGF-1 increases[96] the activity of the 25-hydroxyvitamin D 1α-hydroxylase (“growth factors” in Fig. 7.15). FGF-23 induces renal phosphate wasting in patients with tumor-induced osteomalacia (TIO),[427,439–441] autosomal dominant hypophosphatemic rickets (ADHR), X-linked hypophosphatemic rickets (XLH), and autosomal recessive hypophosphatemic rickets (ARHR).[425,430,432,442] From a physiologic perspective, it would be appropriate for FGF-23 and sFRP-4 concentrations to be regulated by the intake of dietary phosphorus and by serum phosphate concentrations. In humans, in the short-term the feeding of meals containing increased amounts of phosphate does not increase serum FGF-23 concentrations despite the induction of a robust and dose-dependent phosphaturia.[422,443] Other human studies conducted over a period of days or weeks, however, have shown changes in serum FGF-23 concentrations following alterations in the content of phosphate in the diet.[444,445] In mice, Perwad et al. have shown that a high-phosphate diet increased and a low-phosphate diet decreased serum FGF-23 levels in these animals within 5 days of a changing dietary phosphate intake.[446] The changes in serum FGF-23 correlated with changes in serum phosphate concentrations. Studies from our laboratory performed in rats fed a low-, normal, or high-phosphate diet demonstrate that serum FGF-23 levels significantly decrease in animals fed a low-phosphate diet, and increase in animals fed a high-phosphate diet within 24 hours of altering dietary phosphate intake but do not correlate with serum phosphate in the animals fed a high-phosphate diet.[98]

REABSORPTION OF PHOSPHATE ALONG THE NEPHRON

Virtually all serum phosphate is filtered at the glomerulus.[447] Under conditions of normal dietary phosphate intake, and in the presence of intact parathyroid glands, approximately 20% of the filtered phosphate load is excreted. The other 80% of the filtered load of phosphate is reabsorbed by the renal tubules. The PTs are the major sites of phosphate reabsorption along the nephron (Fig. 7.16).[447] Little phosphate reabsorption occurs between the late PT and the early distal tubule in animals with intact parathyroid glands.[448–456] In the absence of PTH, however, phosphate is avidly reabsorbed between the late PT and early distal tubule, reflecting phosphate reabsorption by the proximal straight tubule.[451] Phosphate transport rates are approximately three times higher in the proximal convoluted than in the proximal straight tubules.[457] Renal phosphate handling is characterized by intranephronal heterogeneity, reflecting segmental differences in phosphate handling within an individual nephron as well as internephronal heterogeneity.[448,452,457,458]

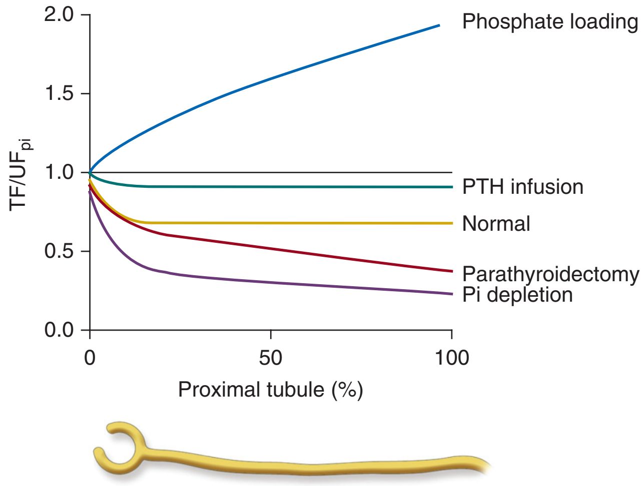


Fig. 7.16 The proximal tubule is the major site of phosphate reabsorption along the nephron. The effects of dietary phosphate loading or deprivation, parathyroid hormone *(PTH)* infusion, or parathyroidectomy on phosphate absorption along the proximal tubule are shown. *Proximal tubule %,* Distance along the proximal tubule as a percentage of total length; *PTH,* parathyroid hormone; *TF/UFPi*, ratio of tubular fluid-to-ultrafiltrate phosphate concentration.

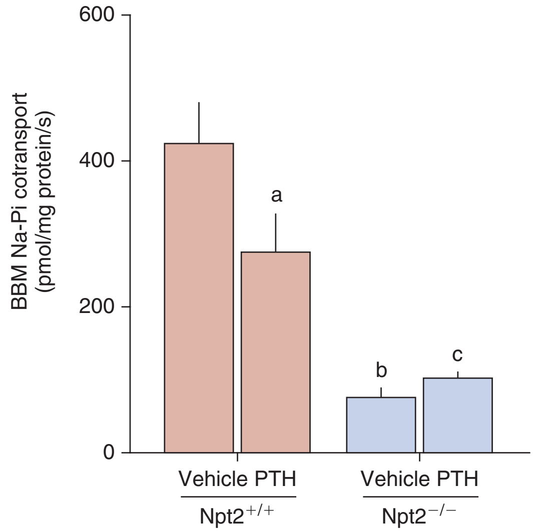


Fig. 7.17 Mice with ablation of the *NaPi-IIa* gene exhibit renal phosphate wasting and fail to respond to parathyroid hormone (PTH). The effect of PTH or vehicle on brush border membrane (BBM) Na-Pi cotransport in *Npt2*$^{+/+}$ and *Npt2*$^{-/-}$ mice is shown. [a]Effect of PTH in *Npt2*$^{+/+}$ mice, $P < .0015$. [b]Effect of genotype in vehicle-treated mice, $P < .0001$; [c]Effect of genotype in PTH-treated mice, $P < .0041$. Reproduced from Zhao N, Tenenhouse HS. Npt2 gene disruption confers resistance to the inhibitory action of parathyroid hormone on renal sodium-phosphate cotransport. *Endocrinology* 2000;141:2159–2165.

The uptake of phosphate is mediated by Na–phosphate cotransporters located at the apical border of PT cells (NaPi–IIa/Slc34A1 and NaPi IIc/Slc34a3).[459–482] The structure and physiology of these phosphate transport molecules have been extensively reviewed, and the reader is directed to other publications in this regard.[459–482] The Na–phosphate cotransporters are highly homologous and are predicted to have similar structures. Mice with ablation of the *NaPi-IIa/Slc34a1* gene exhibit renal phosphate wasting and reduced PT brush border membrane vesicle phosphate uptake (Fig. 7.17).[483] In humans, *SLC34A1* mutations are associated with hypophosphatemia and urinary phosphate losses with urolithiasis or bone demineralization.[484] It is estimated that the NaPi-IIa transporter is responsible for approximately 85% of proximal tubular phosphate transport and contributes to the adaptive increases in tubular phosphate transport in animals fed a low-phosphate diet.[483,485] Mice with constitutive or renal-specific deletions of the *NaPi-IIc/Slc34a3* gene do not display abnormalities in phosphate excretion or phosphate serum concentrations.[486,487] This is in contrast to humans, in whom *SLC34A3* mutations are associated with hypophosphatemic rickets with hypercalciuria.[488,489] The extrusion of phosphate at the basolateral membrane of the proximal tubular cell may be mediated by the xenotropic and polytropic retroviral receptor (Xpr1).[490] Mice with conditional deletions of *Xpr1* exhibited tubular dysfunction with glycosuria, amino-aciduria, hypercalciuria, and phosphaturia and developed hypophosphatemic rickets. In primary cultures of proximal tubular cells, Xpr1 deficiency significantly reduced phosphate uptake and decreased the expression of NaPi-IIa and NaPi-IIc cotransporters.

Interactions between the NaPi-IIa/Slc34A1 and the intracellular protein, the sodium-hydrogen exchanger regulatory factor-1 (NHERF-1), modulate the amount of NaPi-IIa/Slc34A1 and NaPi IIc/Slc34a3 present on the surface of the proximal tubular cell.[491–510] By binding to the NaPi-IIa/Slc34A1 protein, NHERF-1 functions to retain the NaPi-IIa/Slc34A1 protein on the surface of the proximal tubular cell; phosphate uptake diminishes as a consequence of endocytosis of the NaPi-IIa/Slc34A1 when it dissociates from NHERF-1, a process that is activated by the hormonal induction of protein kinase C and the phosphorylation of specific serine and threonine residues on the PDZ domain of NHERF-1.[495,500,506,511–514] Ezrin, a protein that facilitates the association of NHERF-1 to the actin cytoskeleton, also plays a role in the regulation of proximal tubular phosphate transport and the expression of NaPi-IIa/Slc34A1 in proximal tubular cells.[515,516] Ezrin knockdown mice exhibit hypophosphatemia and osteomalacia and a reduction in NaPi-IIa/Slc34A1 and NHERF-1 expression at the apical membrane of PTs. Cellular events associated with uptake of Pi into the cell and extrusion of Pi out of the cell are shown in Fig. 7.18.

REGULATION OF PHOSPHATE TRANSPORT IN THE KIDNEY

DIETARY PHOSPHATE

The influence of dietary phosphate intake on the urinary excretion of phosphate has been known for many years.[448,517–532] The reabsorption of phosphate is decreased in animals fed a high-phosphate diet, whereas animals with a low intake of phosphate reabsorb almost 100% of the filtered load of phosphate.* These changes in phosphate reabsorption are associated with parallel changes in the abundance of NaPi-IIa and IIc.[535,536] In infants and children, phosphate reabsorption is high so as to maintain a positive phosphate balance required for growth.[537,538] Conversely, decreased phosphate reabsorption has been demonstrated in the elderly.[539]

*References 182, 219–228, 234, 240, 241, 354, 533, and 534.

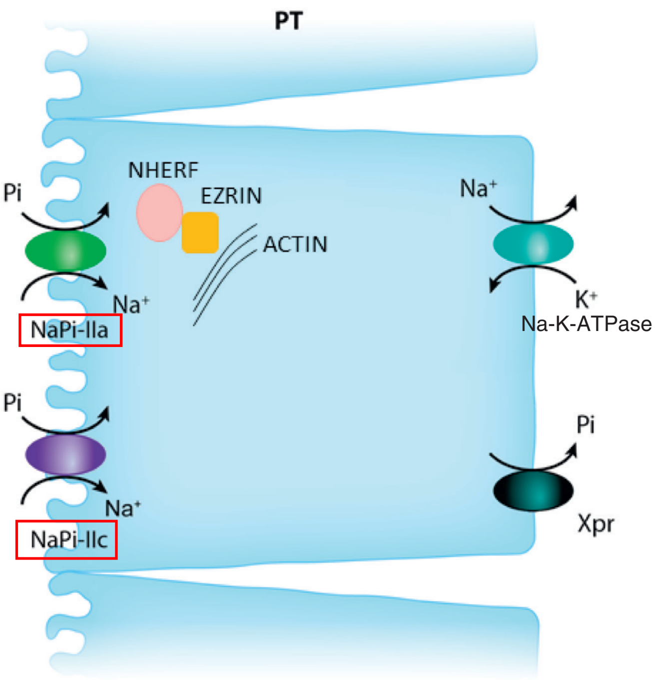


Fig. 7.18 Mechanisms by which phosphate is transported across the proximal tubular cell. Apical uptake is mediated by the sodium-phosphate cotransporters IIa and IIc. The Na-K ATPase located at the basolateral aspect of the cell provides a sodium gradient within the cell that permits these sodium-phosphate cotransporters to take up sodium and phosphate. By binding to the NaPi-IIa protein, NHERF-1 retains the NaPi-IIa protein on the surface of the proximal tubular cell thereby enhancing phosphate uptake; phosphate uptake diminishes as a consequence of endocytosis of the NaPi-IIa when it dissociates from NHERF-1, a process that is activated by the hormonal induction of protein kinase C and the phosphorylation of specific serine and threonine residues on the PDZ domain of NHERF-1. Ezrin, a protein that facilitates the association of NHERF-1 to the actin cytoskeleton also regulates proximal tubular phosphate transport. Ezrin knock down mice exhibit hypophosphatemia and osteomalacia and a reduction in NaPi-IIa/Slc34A1 and NHERF-1 expression at the apical membrane of proximal tubules. The xenotropic and polytropic retroviral receptor (Xpr1) plays a role in the extrusion of phosphate at the basolateral cell surface. *PT,* Proximal tubule.

Although dietary phosphate deprivation and excess results in marked changes in the plasma concentrations of several hormones (see Fig. 7.15) that contribute to the increase or decrease in renal phosphate reabsorption, acute changes in tubular reabsorption can also be demonstrated independent of changes in these hormones.[421,540–543] When a bolus of phosphate is instilled into the duodenum of intact rats, renal phosphate excretion increases within 10 minutes without changes in serum Pi concentrations.[421] The change in Pi reabsorption in response to a high-Pi meal is independent of plasma Pi concentrations and filtered Pi load. Such changes are not elicited upon the administration of NaCl into the duodenum. The increase in renal phosphate excretion is independent of PTH as thyro-parathyroidectomy does not alter the process. Serum concentrations of PTH do not change, and serum concentrations of other phosphaturic peptides, such as FGF-23 and sFRP-4, are unchanged following the infusion of intraduodenal phosphate. Aqueous duodenal extracts contain a phosphaturic substance that is likely to be a protein. The processes or pathways by which changes in luminal phosphate concentrations within the bowel are detected have not been defined, although the presence of a "phosphate sensor" has been postulated.[544] A recent study, though, suggests that such an intestinal phosphate-sensing mechanism may be absent in humans.[545]

Studies using cultured renal proximal tubular cells provide evidence of an intrinsic ability of these cultured cells to increase phosphate transport when exposed to a low phosphate concentration in the medium.[540–543] The mechanism of upregulation of Na/Pi cotransport in OK cells by low-Pi media involves two regulatory mechanisms: an immediate (early) increase (after 2 hours) in the expression of Na/Pi cotransporter, independent of mRNA synthesis or stability, and a delayed (late) effect (after 4–6 hours), resulting in an increase in NaPi-4 mRNA abundance.[542,546] The enhanced Pi reabsorption of short-term Pi deprivation has been linked to decreased intrarenal synthesis of dopamine and/or stimulation of beta adrenoreceptors, because infusion of dopamine or propranolol restores the phosphaturic response to PTH in short-term (less than 3 days) Pi deprivation.[547–549] Conversely, dopamine may also mediate the acute phosphaturic effect of a high-Pi diet.[499] The NaPi-IIa transporter is expressed in the brain and is regulated by dietary Pi, suggesting that dietary Pi could regulate neural outputs and regulate renal Pi excretion.[550] Increasing cerebrospinal fluid Pi concentrations in the presence of low plasma Pi concentrations reversed the adaptations to feeding a low-Pi diet, suggesting that the Pi concentration in the brain regulates not only central but also renal expression of NaPi-IIa transporters. It should be remembered that alterations in serum Pi concentrations also alter $1\alpha,25(OH)_2D_3$ synthesis and serum concentrations.[88,92,94,99–105] Infusions of $1\alpha,25(OH)_2D_3$ increase the renal reabsorption Pi, predominantly in the proximal nephron.[68,551–557]

PARATHYROID HORMONE

Parathyroidectomy decreases renal Pi excretion and, conversely, injection of PTH increases urinary Pi excretion[558–562] by altering Pi reabsorption along the PT (see Fig. 7.16).[450–454,563] The proximal straight tubule is an important site of PTH action with respect to Pi transport and may be critical in the final regulation of Pi excretion.[449,455,458,564] PTH maintains Pi homeostasis by regulating NaPi cotransporters in the kidney. This is mediated by PTH/PTHrP receptors on the apical membrane that signal through protein kinase C (PKC), and on the basolateral membrane that signal through both cAMP/PKA and PKC. This leads to endocytosis of NaPi-IIa and -IIc cotransporters, which are degraded within the lysosomes.[469,474,565,566] The transporters are reduced in number along the apical borders of proximal tubular cells following the administration of PTH 1-34 but not by the administration of PTH 3-34.[479,480] Disruption of the *NaPi-IIa* (*Slc34a1*) gene in mice resulted in increased Pi excretion compared with wild-type mice and a resistance to the phosphaturic effect of PTH (see Fig. 7.17), although the cyclic adenosine monophosphate (cAMP) response is normal.[567] It has been proposed that the primary mechanism for PTH action is PKC-mediated phosphorylation of a PDZ domain on NHERF-1. This leads to dissociation of NaPi-IIa/NHERF-1 complexes, freeing NaPi-IIa from the apical surface for internalization.[500,506]

Under certain conditions, the phosphaturic effect of PTH is blunted or absent. These include short-term Pi deprivation or acute respiratory alkalosis. In these situations, the inhibitory effect of PTH on Pi reabsorption by the proximal convoluted tubule remains intact. However, the increased delivery of Pi leads to enhanced downstream reabsorption by the proximal straight tubule.[455,458,564] These studies suggest that the regulation of Pi reabsorption by PTH in the proximal convoluted and proximal straight tubules may be mediated by different mechanisms. It should be noted that PTH has two opposing effects: PTH increases urinary Pi excretion but also increases the synthesis of $1\alpha,25(OH)_2D_3$ by stimulating the activity of the $25(OH)D_3$ 1α-hydroxylase enzyme in the kidney.[88,92,94,99–105]

VITAMIN D AND VITAMIN D METABOLITES

Dietary Pi deprivation or hypophosphatemia induces 25-hydroxyvitamin D_3 1α-hydroxylase.[88,92,94,99–105] Mice or rats, but not pigs, fed a low-Pi diet show a decrease in the activity of the 25-hydroxyvitamin D_3 24-hydroxylase (a renal enzyme involved in the catabolism of $1,25(OH)_2D_3$) compared with rats fed a normal Pi diet within 24 hours of Pi restriction.[83,568,569] $1,25(OH)_2D_3$ decreases renal Pi excretion,[62,68,551–554] but the mechanism remains unknown. VDR-mutant mice exhibit decreased serum Pi; however, Pi transport or by renal cortical brush border membranes, Pi excretion, or NaPi-IIa or NaPi IIc mRNA levels were not different between VDR-null or wild-type mice, whereas NaPi-IIa protein expression and NaPi-IIa cotransporter immunoreactive signals were slightly but significantly decreased in the VDR mice compared with the wild-type mice.[536] When VDR knockout mice were fed a low-Pi diet, serum Pi concentrations were more markedly decreased in the VDR knockout mice than in the wild-type mice. Other studies performed in VDR and 25(OH)D 1α-hydroxylase–null mutant mice show that both these knockout mice adapt to Pi deprivation with increased NaPi-IIa protein in a manner similar to that found in wild-type mice.[570] However, when these mice were fed a high-Pi diet, Pi excretion was less in the VDR and 25-hydroxyvitamin D 1α-hydroxylase–null mutant mice compared with the wild-type mice. In vitamin D–deprived rats, NaPi-IIa transporter protein and mRNA were reported to be decreased in juxtamedullary but not superficial renal cortical tubules compared with normal rats.[571]

INSULIN, GROWTH HORMONE, AND INSULIN-LIKE GROWTH FACTOR

Insulin decreases plasma Pi and Pi excretion in human and animal models.[572–575] This enhanced renal Pi reabsorption can be demonstrated in the absence of changes in blood glucose, PTH, and Pi levels or urinary Na excretion. Micropuncture studies[573] demonstrate enhanced Pi reabsorption in hyperinsulinemic dogs and somatostatin infusion, which decreases plasma insulin levels and increases Pi excretion.[576] Growth hormone decreases Pi excretion and has been postulated to contribute to increased Pi reabsorption and positive Pi balance demonstrated in growing animals.[577,578] Administration of a growth hormone antagonist for 4 days to immature rats is associated with increased Pi excretion and a decreased transport capacity for Pi reabsorption.[579,580] In juvenile rats suppression of growth hormone is associated with an increase in Pi excretion as a result of decreased NaPi-IIa expression.[581] Growth hormone administration increases Pi uptake by brush border membrane vesicles.[582] Because growth hormone increases renal insulin-like growth factor-1 (IGF-1 synthesis),[583] the effects of growth hormone on Pi reabsorption may also be due to IGF-1.[577,583–588]

RENAL NERVES, CATECHOLAMINES, DOPAMINE, AND SEROTONIN

Numerous studies have demonstrated that acute renal denervation or the administration of catecholamines alters Pi reabsorption regardless of PTH.[547,589–601] The increase in urinary Pi excretion after acute renal denervation could be due to both increased production of dopamine and decreased α- or β-adrenoreceptor activity, because acute renal denervation has been shown to initially increase renal dopamine excretion and almost completely abolish renal norepinephrine and epinephrine levels.[602,603] Epinephrine decreases plasma Pi, presumably by shifting Pi from the extracellular into the intracellular space. The hypophosphatemic response to isoproterenol infusion is blocked by propranolol, suggesting involvement of the beta adrenoreceptors. Infusion of isoproterenol markedly enhances renal Pi reabsorption in normal rats and in hypophosphatemic mice.[600,604] The enhanced Pi reabsorption and attenuated phosphaturic response to PTH observed in acute respiratory alkalosis and Pi deprivation is blocked by infusion of propranolol, suggesting a possible role for stimulation of β-adrenoreceptors in these conditions. Stimulation of α-adrenoreceptors by the addition of epinephrine to OK cells blunts the PTH-induced increase in cAMP levels and the inhibition of Pi transport.[605] Stimulation of α2-adrenoreceptors in vivo has also been demonstrated to attenuate the phosphaturic response to PTH.[548] Dopamine infusion and the infusion of L-dopa or gludopa, or dopamine precursors, increase Pi excretion in the absence of PTH.[606–608] Dopamine administration decreases Pi transport in OK cells and rabbit proximal straight tubules.[599,609–614] Increasing dietary Pi intake increases urinary dopamine excretion and Pi excretion.[615] Inhibition of endogenous dopamine synthesis by the administration of carbidopa to rats results in decreased dopamine and Pi excretion, suggesting a role for endogenous dopamine in Pi regulation.[595,603] A paracrine role for dopamine in Pi regulation is strengthened by studies in OK cells showing that the addition of dopamine or L-dopa selectively decreases Pi uptake. Furthermore, Pi-replete OK cells produce more dopamine from L-dopa than Pi-deprived cells.[611] Dopamine inhibits Pi transport by multiple mechanisms, including activation of DA1 and DA2 receptors.[610,613,614] Dopamine induces the internalization of NaPi-IIa cotransporter molecules by activation of luminal DA1 receptors.[609] Renal PTs also synthesize serotonin from 5-hydroxytryptophan using the same enzyme that converts L-dopa to dopamine. Incubation of OK cells with either serotonin or 5-hydroxytryptophan enhances Pi transport and raises the possibility that serotonin may also be involved in the physiologic regulation of renal Pi transport.[606,612,616,617]

PHOSPHATOMINS (FGF-23, sFRP-4)

The term “phosphatonin” was introduced to describe a factor or factors responsible for the inhibition of renal phosphate reabsorption and altered 25(OH)D 1α- hydroxylase regulation observed in patients with tumor-induced osteomalacia.[440] Cai et al.[439] described a patient with TIO in whom the biochemical characteristics of hypophosphatemia, renal phosphate wasting, and reduced serum 1α,25(OH)$_2$D disappeared following removal of the tumor. Several factors have been identified that are associated with phosphate wasting, including FGF-23 sFRP-4, fibroblast growth factor 7 (FGF-7), and matrix extracellular phosphoglycoprotein (MEPE).

The most extensively studied phosphatonin is FGF-23, a 251–amino acid–secreted protein.[419,425,431,618] Recombinant FGF-23 administered intraperitoneally to mice or rats induces phosphaturia and inhibits 25-hydroxyvitamin D 1α-hydroxylase activity.[419,425,431,618] The minimal sequence needed for phosphaturic activity resides between amino acids 176 and 210.[431] Transgenic animals overexpressing FGF-23 are hypophosphatemic, phosphaturic, and show the presence of rickets and reduced serum 1α,25(OH)$_2$D concentrations or 25-hydroxyvitamin D 1α-hydroxylase activity.[433,434,619] Conversely, mice in which the *FGF-23* gene has been ablated demonstrate hyperphosphatemia, reduced phosphate excretion, markedly elevated serum 1α,25(OH)$_2$D concentrations and renal 25-hydroxyvitamin D 1α-hydroxylase mRNA expression, vascular calcification, and early mortality.[434,620] The ablation of the VDR in FGF-23–null mice has been reported to rescue this phenotype, supporting an important role for vitamin D in the pathogenesis of the abnormal phenotype seen in FGF-23–null mice.[621]

FGF-23 binds and signals through FGF receptors 1c, 3c, and FGFR4[269]; the role of Fgfr3 and Fgfr4 has not been established in mice in vivo.[622] Han et al. recently demonstrated that mice with deletion of *Fgfr1* in the PT had an increase in sodium-dependent phosphate cotransporter expression, hyperphosphatemia and refractoriness to the phosphaturic action of FGF-23, suggesting that FGFR1c plays a key role in its effects in the PT.[623] Deletion of the *Fgfr1* in the distal tubule resulted in hypercalciuria and secondary hyperparathyroidism. A coreceptor, klotho, is necessary for FGF-23 to exhibit bioactivity.[269,624] The role of klotho in FGF-23 signaling is supported by the observation that klotho knockout mice have a phenotype identical to that of FGF-23 knockout mice,[625] whereas a human mutation that increases klotho levels phenocopies TIO and X-linked hypophosphatemic rickets.[626]

The mechanism for FGF-23 action is thought to involve downstream signaling through ERK1/2 and serum and glucocorticoid kinase-1 (SGK1).[627] SGK1 in turn phosphorylates NHERF1, leading to the dissociation of NaPi-IIa transporters that become internalized and degraded, analogous to its regulation by PTH.[628] This model is supported by the observation that FGF-23 is no longer phosphaturic when given to NHERF1-null mice.[629] Recent evidence suggests that Jak3 may also be involved because Jak3-null mice have renal phosphate wasting and elevated FGF-23 levels.[630]

FGF-23 synthesis is regulated by 1α,25(OH)$_2$D. Increasing doses of 1α,25(OH)$_2$D increase FGF-23 concentrations in the serum within 24 hours, but statistically significant changes are observed 4 hours after 1α,25(OH)$_2$D treatment.[631,632] In the physiologic sense, it is possible that FGF-23 is a negative feedback regulator of the 25-hydroxyvitamin D 1α-hydroxylase enzyme.

The Wnt antagonist, sFRP-4, is highly expressed in tumors associated with renal phosphate wasting and osteomalacia.[429] Recombinant sFRP-4 is phosphaturic in rats and prevents the upregulation of the 25-hydroxyvitamin D 1α-hydroxylase enzyme seen in the presence of hypophosphatemia.[424] sFRP-4 decreases Na^+-Pi cotransporter abundance in the brush border membrane of the PT, and reduces the surface expression of the Na^+-Pi-IIa cotransporter in PTs of the kidney, as well as on the surface of OK cells.[438] sFRP-4 expression is increased in bone samples and serum from X-linked hypophosphatemic mice in mice with a global knockout of the *phex* gene but not in mice in which the *phex* gene has been knocked out in bone alone.[633] sFRP-4 protein concentrations are increased in the kidneys of rats fed a high-phosphate diet for 2 weeks but not in animals fed a low-phosphate diet, suggesting a possible role for sFRP-4 during increases in phosphate intake.[634] This suggests in turn that sFRP-4 concentrations are altered in the kidney of animals fed a high-phosphate diet and could play a role in the long-term adaptations to high-phosphate intake.

MEPE is abundantly overexpressed in tumors associated with renal phosphate wasting and osteomalacia.[635] Recombinant MEPE is phosphaturic and reduces serum phosphate concentrations when administered to mice in vivo.[636] The protein has been shown to inhibit phosphate reabsorption in the proximal convoluted tubule,[637] to inhibit Na-dependent phosphate uptake in opossum kidney cells, and to reduce PT expression of NaPi-IIa protein.[638] The protein has also been demonstrated to reduce intestinal Pi absorption directly.[638] MEPE also inhibits bone mineralization in vitro, and MEPE-null mice have increased bone mineralization.[639] Thus, it is possible that MEPE is important in the pathogenesis of hypophosphatemia in renal phosphate wasting observed in patients with TIO. However, MEPE infusion does not recapitulate the defect in vitamin D metabolism seen in patients with TIO.[636] Infusion of MEPE reduces serum phosphate concentrations, and serum 1α,25(OH)$_2$D concentrations increase following MEPE as would be expected in the face of hypophosphatemia. Thus, in patients with TIO, it is likely that MEPE contributes to the hypophosphatemia, but other products such as FGF-23 and sFRP-4 inhibit 1α,25(OH)$_2$D concentrations by inhibiting the activity of the 25-hydroxyvitamin D 1α-hydroxylase. MEPE may play a role in the pathogenesis of X-linked hypophosphatemic rickets, in which there is phosphate wasting, and evidence for a mineralization defect that is independent of low phosphate concentrations in the extracellular fluid.[633] MEPE expression is increased in mice with the *Hyp* mutation, and mice with a global knockout of the *phex* gene but not in mice with a bone specific knockout of the *phex* gene. It is not known whether MEPE is regulated by phosphate concentrations although Jain et al. have demonstrated that it is correlated with serum Pi concentration in normal humans.[640] Another growth factor, FGF-7, also known as keratinocyte growth factor, is overexpressed in tumors associated with phosphate wasting and osteomalacia.[488] FGF-7 inhibits Na-dependent phosphate transport in OK cells, and we have demonstrated that FGF-7 inhibits renal phosphate reabsorption in vivo. FGF-7 is present in normal plasma and is significantly increased in patients with renal failure (personal observations). Whether or not FGF-7 is regulated by phosphate concentrations is unknown.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. A 35-year-old woman with intermittent diarrhea of 6 months' duration is referred to you for evaluation of bone pain. Previous evaluation has shown the presence of celiac disease. The following tests are available for evaluation of her symptoms.

Hemoglobin 10.5 g/dL, mean corpuscular volume 105 fL, leukocyte count 6.5×10^9/L, platelet count 165×10^9/L, serum folate 2.5 µg/L, serum sodium 135 mEq/L, potassium 3.7 mEq/L, chloride 105 mEq/L, bicarbonate 22 mEq/L, creatinine 0.9 mg/dL, blood urea nitrogen 12 mg/dL, total serum calcium 8.5 mg/dL, ionized calcium 4.0 mg/dL, phosphorus 2.5 mg/dL, alkaline phosphatase 200 U/L, parathyroid hormone 80 pg/mL, 25-hydroxyvitamin D 10 ng/mL, and 1,25-dihydroxyvitamin D 30 pg/mL. Urinalysis is normal. Urine microscopy is normal. Bone densitometry shows osteopenia.

A 24-hour urine sample is collected for measurement of calcium, phosphorus, and creatinine. In regard to her urinary analytes, one will anticipate that (select all correct answers):

a. 24-hour urine calcium will be low.
b. 24-hour urine calcium will be high.
c. 24-hour urine phosphorus will be low.
d. 24-hour urine phosphorus will be high.
e. None of the above

Answers: a and d

a. Fractional excretion of calcium will be low.
b. Fractional excretion of calcium will be high.
c. Fractional excretion of phosphorus will be low.
d. Fractional excretion of phosphorus will be high.

Answers: a and d

Rationale: The patient has secondary hyperparathyroidism due to intestinal malabsorption of calcium. One would anticipate low urinary calcium excretion and a low fractional excretion of calcium as a result of elevated parathyroid hormone concentrations One would expect urinary phosphorus to be elevated and fractional excretion of phosphorus to be high on account of secondary hyperparathyroidism.

2. The major effect of parathyroid hormone (PTH) on urinary calcium reabsorption occurs in the

a. Proximal tubule
b. Thick ascending limb of the loop of Henle
c. Distal convoluted tubule
d. Collecting duct
e. All of the above

Answers: b and c

Rationale: The major sites of parathyroid hormone–mediated calcium absorption are in the thick ascending limb and the distal convoluted tubule. There is no effect of PTH on calcium absorption in the proximal tubule.

3. A 35-year-old female is referred to you for evaluation of bone pain. She was completely well 2 years ago when she started developing intermittent aches and pains. Family history is noncontributory. Examination is negative except for a 1 × 2 cm subcutaneous mass in the left popliteal fossa.

Laboratory values show the following: hemoglobin 12.8 g/dL, MCV 85 fL, leukocyte and platelet count. Serum sodium, potassium, chloride, and bicarbonate are normal. Serum calcium is 9.2 mg/dL, ionized calcium 4.6 mg/dL, phosphorus 2.0 mg/dL, alkaline phosphatase 150 U/L, 25-hydroxyvitamin D 22 ng/mL, 1,25-dihydroxy vitamin D 25 pg/mL, and PTH 30 pg/mL. Which one of the following tests will assist in making the diagnosis?

a. Serum FGF-23
b. 24-hour urine phosphorus and creatinine
c. Bone density
d. Sestamibi technician scan
e. Magnetic resonance scan of the left tibial fossa
f. All of the above

Answer: f

Rationale: The patient has tumor-induced osteomalacia based on the presence of a low serum phosphorus, normal serum 25-hydroxyvitamin D concentrations, and a low-normal 1,25-dihydroxyvitamin D and PTH concentration. Investigation will reveal an elevated 24-hour urine phosphorus and an increased fractional excretion of phosphorus. TmP/GFR will be suppressed. FGF-23 concentrations will be increased. Bone density will be reduced. A sestamibi technician scan will show uptake over the popliteal fossa. This will be confirmed by magnetic resonance spectroscopy.